# Chapter 11

# Other conditions

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# Myxoma

## Key facts

- Myxomas are composed of pluripotential ***subendocardial mesenchymal*** cells (📖 p84) and are the commonest type of primary cardiac tumor, compromising 40–50% of all benign cardiac tumors.
- They are 2–3× commoner in women, occurring in the 3–6th decade. 5% show autosomal dominant inheritance.
- 70% of sporadic myxomas occur in the LA.
- They are an indication for urgent removal, to reduce the risk of systemic embolism

## Presentation

Symptoms and signs depend on the location of the myxoma and the degree to which RV and LV filling is compromised. Often incidental finding in asymptomatic patient. Embolic and constitutional features coexist.

- LA myxoma (75%): episodic dyspnea, hemoptysis, syncope.
- RA myxoma (20%): episodic dyspnea, signs of right heart failure, ascites and hepatic congestion, ↑JVP, edema.
- ***Embolic features*** (30–40%): mostly systemic, ***stroke***, retinal, visceral, coronary, aortic saddle embolus, PE rare.
- Very large mobile tumors may transiently occlude AV valve leading to loss of cardiac output (which can occasionally be restored by placing patient in steep Trendelenburg to dislodge tumor) and ***sudden death***.
- Systolic murmurs (impeded AV closure/narrowing outflow tracts).
- Diastolic murmurs (obstructed filling of RV or LV).
- Widely split S2 (delayed closure of mitral valve), 'tumor plop'.
- Constitutional features: fatigue, fever, rash, pyrexia, athralgia, myalgia, ↓weight, ↓Hb, ↓Plts, ↑CRP, ↑ESR.

## Investigation

- EKG changes are non-specific and include AF and bundle branch block.
- CXR changes are non-specific: calcification is unusual, signs of LA enlargement and pulmonary edema may feature.
- Echocardiography is diagnostic: TEE can detect masses as small as 1–3mm and a ***pedunculated, smooth mobile mass sitting in the LA*** almost pathognomonic of LA myxoma.



## Natural history

Myxomas are usually benign, but the tumor occasionally metastasizes to brain, sternum, vertebra, pelvis, or scapula. Once myxomas become symptomatic death usually occurs within 1–2 years. There is a significant risk of sudden death: 10% of pre-op patients die of embolic phenomena.

- Hospital mortality after resection of atrial myxomas is about 1–5%.
- Hospital mortality after resection of ventricular myxomas is about 10%.
- 1–2% of sporadic myxomas recur, compared to up to 75% of familial.

## Indications for operation

Myxomas have been traditionally considered as an indication for urgent surgery (within 24h of diagnosis).

## Surgical technique for left atrial myxoma

Minimize handling the heart to avoid fragmentation of the tumor. In the rare event of sudden loss of cardiac output before bypass is instituted as a result of a large tumor occluding flow into the LV, place the patient in steep Trendelenburg to allow the tumor to roll back into the LA restoring output, start chest compressions if needed, and open emergently.

***Set-up***

- Standard invasive monitoring including TEE.
- Median sternotomy (right anterolateral thoracotomy an option).
- Incise the pericardium away from the midline so that you can harvest a patch for repair of large defects.
- Insert a mitral retractor hitching up the right pericardium only.
- Standard aorto-bicaval cannulation with snares and/or vacuum assist, $CO_2$ insufflation, and an aortic root vent.
- Surgery is performed with the heart fibrillating, or clamped with antegrade cardioplegic arrest at 32°C.
- Pump suckers, vents, and cell salvage may be used as they add little to the risk of embolizing tumor cells arising from bypass.

***Atriotomy and tumor excision***

- Atriotomy is dictated by the location of the tumor which is usually attached to the left side of the interatrial septum: this is best approached via an *oblique right atriotomy*, then incising the floor of the right atrium in a *transseptal approach*
- Exposure is gained as in a tricuspid repair, with retractors and by extending the incision if appropriate (📖 p404).
- The tumor is most commonly attached by a stalk, which may be broad based, to the atrial septum: this base should be excised with a *5mm margin* of full thickness septum.
- A sterile *dessert spoon* is useful to help evacuate the friable, gelatinous tumor without fragmenting it.
- Once the tumor is removed inspect the whole of the LA, the LV, and what can be seem of the RA through the septal defect, and RV through the tricuspid valve, for any other myxomas, which should be excised.
- Ventricular myxomas are excised with a partial thickness of ventricular wall: the approach is usually via the LA and RA for LV and RV tumors respectively.
- Occasionally myxomas arise from valves, or damage valves by a 'wrecking ball' effect which requires reconstruction or replacement.
- Thoroughly irrigate with water (may lyse residual tumor cells) and a waste sucker (not cell salvage).

***Closure***

- Repair small septal defects (<1–2cm) with double layered 4/0 running Prolene, and larger defects with the pericardial or Dacron patch, de-airing the LA by asking the anesthesiologist to ventilate, displacing air from the LA as the suture line is completed (📖 p120).
- Place RV pacing wires as tumor excision can compromise SA node artery which crosses the interatrial septum, leading to sick sinus syndrome.

# Other cardiac neoplasms

## Key facts

Pathology and indications for operation of primary cardiac neoplasms, including lipomas, papillary fibroelastomas, rhabdomyomas, angiosarcomas, and rhabdomyosarcomas, are outlined on 📖 p84. These tumors are removed by open heart surgery where possible.

## Right atrial spread of infradiaphragmatic tumors

***Pathology***

Infradiaphragmatic tumors that extend up the IVC into the RA normally involve both the endothelium of the IVC and the endocardium of the RA. Tumors include:

- *Renal cell carcinoma* (by far the commonest).
- Renal adenocarcinoma.
- Wilms tumor.
- Uterine leiomyosarcoma.

***Diagnosis***

- Cardiac symptoms do not occur until quite late. Extension into the right atrium may be suspected from CT imaging performed in order to stage tumors, and can be further assessed by TEE or MRI.

***Indications for surgery***

Renal cell carcinoma extension without distant metastasis is one indication, although indications in less common tumors are less well established. After combined nephrectomy and resection of atrial renal cell carcinoma:

- 5-year survival is ~65%.
- 10-year survival is ~60%.

***Surgical technique***

The nephrectomy and resection of atrial extension are most commonly performed as a combined procedure. It is often possible for the vascular surgeons to complete the entire procedure without cardiopulmonary bypass from the abdomen. Occasionally this is not possible, and *cardiopulmonary bypass* is required for an approach via the RA. As no venous return can be obtained from the IVC while the tumor is being excised *deep hypothermic circulatory arrest or low flows* at deep hypothermia are required.

- A TEE probe is placed in addition to routine invasive monitoring.
- The patient is prepped and draped as for CABG, but exposed from clavicle to groin, with no drapes over the umbilicus.
- A median sternotomy incision is extended to a midline laparotomy as far as the pubis.
- The SVC and RA are cannulated (do not use a two-stage cannula), aortocaval bypass commenced using vacuum assist, and the patient cooled to 18°C, venting either the LV or the right superior pulmonary vein.
- Pump suckers, vents, and cell salvage may be used as they add little to the risk of embolizing tumor cells arising from bypass.
- The abdominal tumor and involved kidney is mobilized ready for excision during cooling.

- At 20°C the aorta is cross-clamped, cold blood cardioplegia given into the root, the SVC snared, and the RA opened.
- Flows can be reduced or stopped for up to 35min at this temperature, usually 5–10min of low flow or DHCA are necessary to clearly visualize the tumor in the IVC orifice.
- A rough sucker is used to remove blood from the vicinity of the tumor in the RA as it is dissected free of the endocardium.
- Blunt dissection to free the tumor from the endothelium of the IVC.
- Once the tumor has been removed from the RA and IVC, the renal pedicle is ligated and divided, and the kidney removed.
- The RA is closed and the IVC cannulated in the normal manner so that full flow bypass can be established and the patient rewarmed.

## Carcinoid heart disease

### *Pathology*

Carcinoid tumors arise from the *argentaffin cells of the GI tract*. They occur in the ileum, pancreas, ovaries, and testes, and they secrete large amounts of serotonin (5HT). 10% metastasize, usually to the liver. 10% result in *carcinoid syndrome* which is characterized by flushing, bronchospasm, diarrhea, and telangiectasia. In 50% of cases of carcinoid syndrome fibrosis involves the right heart valves. Severity of cardiac involvement correlates with serum levels of 5HT. Cardiac involvement results in *tricuspid and/or pulmonary valve regurgitation* due to thickening and restriction of the valve leaflets, and is usually limited to the right-sided valves. Metastasis to the heart is rare.

### *Diagnosis*

The described symptoms may be accompanied by signs including the systolic murmur of TR. While carcinoid syndrome is diagnosed from symptoms, imaging identifying the tumor, serum 5HT and urinary 5-hydroxyindoleacetic acid, cardiac lesions are best assessed by echocardiography. Valve leaflets appear short and thick with a characteristic 'dagger' appearance in color modes.

### *Indications for surgery*

Valve lesions progress rapidly and cause CHF in untreated patients, who have a poorer prognosis than carcinoid patients without cardiac involvement. Operative mortality 10–20%. Liver metastases are not a contraindication to surgery as they are indolent and treatable, but predispose the patients to *carcinoid crises* (profound hypotension), triggered by catecholamine use, and prevented and treated by octreotide infusion which should be maintained throughout the perioperative period.

### *Techniques of surgery*

Valve repair is not usually possible. Pulmonary and tricuspid replacement (📖 p417) is performed, usually with bovine pericardial patch augmentation of the RVOT. Bioprostheses are preferred despite accelerated degeneration if 5HT levels are high, as mechanical valves and anticoagulation make postoperative care and future resection of liver metastases challenging.

# Massive pulmonary embolism

## Key facts

- ***Acute PE*** is common in patients with multiple risk factors, particularly if not on thromboembolic prophylaxis.
- The majority of cases are managed medically with ***heparinization or thrombolysis:*** indications for emergency surgery are unusual but include near complete occlusion of both pulmonary arteries with either evidence of hemodynamic compromise (hypotension, shock, cardiac arrest) or RV dysfunction, or large intracardiac thrombus.

## Pathology

### Risk factors for venous thromboembolism

In 1856 Rudolf Virchow linked DVT and PE to *a triad of venous stasis, vein wall injury, and hypercoagulability*. Known risk factors for DVT include:

***Venous stasis***

- Immobility (surgery, trauma, paralysis, medium- and long-haul flights).
- Failure of deep venous valves, venous dilatation.
- Compression of veins (pregnancy, bulky intra-abdominal tumors and cysts, deck chairs, poor padding on operating table, tourniquets).
- Low cardiac output, right heart failure.

***Endothelial injury***

- Previous DVT, traction on vein branches during surgery.

***Hypercoagulability***

- Dehydration, age, malignancy, smoking, pregnancy.
- Oral contraceptive pill, factor V Leiden, C and S deficiency.

Venous thrombus most commonly forms in the deep veins of the leg: 90% of clinically detected PEs are associated with leg DVT. Thrombus may propagate into the iliac veins or spontaneously detach travelling through the right side of the heart into the PA, either as a single large clot which may lodge straddling the pulmonary trunk (saddle embolus) or as smaller fragments which tend to migrate to the lower lobes, right more than the left. Old-established DVT is usually firmly attached to the vessel walls and less likely to embolize than a new clot which is usually only attached to the vein wall at its site of origin. The thrombus rapidly becomes covered in a layer of platelets and thrombin. As a result of acute PE several pathological changes occur:

- RV pressure rises causing RV dilatation.
- RV stroke volume and consequently LV preload fall.
- Systemic hypotension results in reduced coronary blood flow.
- Alveolar dead space increases causing $\downarrow\downarrow PaO_2$.
- Except in massive PE, hyperventilation often causes $\downarrow PaCO_2$.
- Pulmonary vasoconstriction is caused by release of serotonin, ADP, thromboxane, platelet activating factor and platelet derived growth factor, anoxia, and ischemia further compromising RV function.

## Diagnosis of pulmonary embolism

PE is incorrectly diagnosed in almost 75% of patients. The differential includes acute MI, aortic dissection, septic shock, chest infection, hemothorax, and pneumothorax. *Massive PE* is PE resulting in hemodynamic compromise, or >30% of the pulmonary vasculature is compromised.

### Clinical features of pulmonary embolism

- Symptoms: dyspnea, pleuritic or dull chest pain.
- Signs: tachypnea, tachycardia, hypotension, elevated JVP, pallor.
- Risk factors for or clinical evidence of DVT.
- Bedside tests: EKG shows RV strain pattern (S1, Q3, T3) but this is neither a specific nor a sensitive test.
- ↓$PaO_2$, $PaCO_2$ may be low.
- CXR may show consolidation and effusion early on.
- TTE or TEE shows RV dilatation, TR, and RA or RV thrombus.
- Pulmonary angiography, spiral CT, and VQ scanning are diagnostic.

#### *CT*

This is the most readily available modality in most centres. CT angiography has a sensitivity of 90% and a specificity of 95% for large PEs. Subsegmental PEs are frequently not picked up.

#### *Ventilation-perfusion scans*

A gamma camera images uptake of inhaled krypton-81m and emission of injected technetium 99m. PE shows up as a perfusion cold spot (emission of $^{81}Kr$ but no $^{99}Tc$). Pneumonia, atelectasis and previous PE also cause VQ mismatches so are relative contraindications to a VQ scan. VQ scans have a 98% sensitivity and low specificity in the diagnosis of acute major PE. Results are given as high, intermediate, or low probability of PE.

#### *Pulmonary angiography*

Pulmonary angiography is the gold standard investigation for acute PE. The associated mortality is 0.2%, and it should not be carried out in hemodynamically unstable patients.

#### *MRI*

This has a specificity of 80% and a sensitivity of 77%. It is better at identifying subsegmental PEs than CT scans, but limited availability and high cost mean that MRI is not a routinely used modality.

## Outcome

- The 30-day mortality of acute massive PE is about 50%.
- About 10% of mortality occurs within the first hour.
- Up to 80% of mortality occurs within the first 2h.
- The mortality of surgical intervention is up to 70% for patients requiring CPR or mechanical circulatory support preoperatively.
- The operative mortality of stable patients is about 10–20%.

# Pulmonary embolectomy

## Key facts

- This a surgical emergency: patients may be *in extremis*.
- The embolus is removed on CPB via two arteriotomies in the MPA and RPA, periods of low flow or circulatory arrest may be needed.
- An *IVC filter* should be placed intraoperatively or immediately after.
- There is often severe RV dysfunction postoperatively.

## Management

CPR may be indicated. Otherwise:

- Sit up and give 100% $O_2$.
- Patient may require intubation before arrival in the OR.
- Get iv access and give heparin 5000U bolus iv.
- Start a heparin infusion (50,000U heparin in 50mL of normal saline) at 1000–2000U/h.

***Indications for surgery***

- Patients with a definite diagnosis of acute massive PE and life-threatening circulatory insufficiency or moderate to severe RV dysfunction, in whom thrombolysis is contraindicated or unsuccessful.
- Large right atrial or ventricular thrombus.

***Contraindications***

Advanced age, frailty, or comorbidity, prolonged CPR.

## Technique of pulmonary embolectomy

- Contact *interventional radiology or cardiology* to arrange placement of IVC filter either in OR or in catheter lab after surgery.
- Ask for gallstone forceps: it may take cardiac team a while to locate these from general surgery.

***Set-up***

- Standard invasive monitoring with TEE.
- Midline sternotomy, full heparinization.
- Aorto-bicaval cannulation with caval tapes or vacuum assist (exercise caution when placing the IVC cannula if there is RA thrombus—simply placing it carefully into the RA is sufficient for institution of CPB), and aortic root vent.
- Cardioplegic or fibrillatory arrest.
- Moderate hypothermia (32°C).

***Procedure***

- Evacuate any clot from the RA and RV via a right atriotomy first: snare the cavae if not using vacuum assist and make an oblique right atriotomy, which can be retracted open using a Cosgrove retractor.
- Close the atriotomy using 4/0 Prolene double layer suture.
- The PA is opened longitudinally 1–2cm distal to the pulmonary valve and the incision extended to the left PA (Fig. 11.1).
- Thrombus is removed ideally using gallstone forceps which are a perfect size and shape for this, and come in a range of curvatures

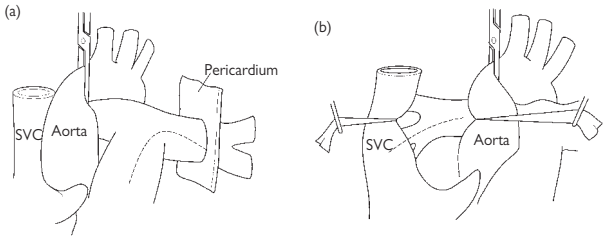



**Fig. 11.1** Pulmonary embolectomy. (a) Left pulmonary arteriotomy (which does not extend past the pericardial reflection). (b) Right pulmonary arteriotomy (which may be even more readily accessed *lateral* to the SVC immediately medial to the pericardial reflection).

which are great for getting around the pulmonary branch arteries. Sponge holders are too straight and large for more distal clots.
- The right PA is opened either between the aorta and the SVC (mobilizing the SVC as far as the innominate and retracting improves exposure) or more easily lateral to the SVC which improves access to more distal branches and is easier to repair.
- Use a nerve root or vein retractor to hold the arteriotomies open.
- A sterile fiberoptic bronchoscope may be used to visualize thrombus in lower level pulmonary vessels, and gently compressing the lung and hilum can help to propel thrombus into the larger pulmonary vessels where it can be suctioned out.
- Take care not to inadvertently grab the PA branch bifurcations instead of clot when extracting blindly as this will cause hemoptysis from localized ruptures postoperatively.
- Aim to extract clot that tapers down to fine point, indicating that it has not broken off in the distal arteries.
- Flush copiously with saline and suction, which improves visualization of remnants of thrombus, and flushes out debris.

***Closure***
- Close the arteriotomies with 4/0 Prolene SH needle, double layer.
- De-air and wean from bypass in the standard fashion: moderate inotopic support for RV dysfunction usually required.
- An IVC filter can be placed percutaneously at this point, or after transfer back to cardiac catheterization labs: some surgeons place it directly under vision while on bypass.
- CVP, RVEDP, and PAP should be much lower than preoperatively.

## Postoperative management
- Heparinization once bleeding stopped (within 6–12h): this is because the pulmonary endothelium may be thrombogenic.
- Start warfarin, and run thrombophilia screen if not already done.

# Chronic pulmonary thromboembolism

## Key facts

- This is a completely different condition, with very different operative approach and outcome to acute massive PE.
- Pulmonary thromboendarterectomy performed for chronic disease is limited to a few centers to try and improve on historically high operative mortality and poor long-term outcomes.

## Pathology

Chronic thromboembolic disease results from either a single acute massive PE that does not resolve, or repeated smaller PEs. There are several reasons why emboli fail to resolve:

- Large volume of clot may overwhelm the thrombolytic cascade.
- If the clot obstructs the vessel lumen, thrombolytics cannot reach it all.
- PEs tend to occur in people with abnormalities of thrombolysis.
- The clot may be made of material other than thrombus that will not resolve, e.g., tumor fragments particularly myxoma, fat, already organized thrombus; or be associated with indwelling lines.

Clot that does not resolve either becomes canalized and endothelialized, or becomes a solid mass of dense fibrous material, with its own capillary blood supply, that completely obstructs the vessel lumen. Pulmonary hypertension results. This is not because of simple obstruction to flow: patients do not develop pulmonary hypertension after pneumonectomy. Sympathetic stimulation and humoral changes contribute. The proximal vasculature becomes dilated and the distal vasculature shows intimal proliferation and medial hypertrophy.

## Presentation

Symptoms and signs are non-specific, and generally do not develop for months to years. Symptoms and signs reflect pulmonary hypertension and RV failure and include exertional dyspnea, fatigue, exertional presyncope, angina, hemoptysis, hypoxia, cyanosis, peripheral edema, ↑JVP, cannon v wave, sternal heave, pansystolic murmur of TR, mid-diastolic murmur of PR.

## Investigation

- EKG shows right heart strain pattern (right axis deviation and dominant R wave in V1).
- CXR may show oligemia or abrupt cut-offs of the hilar vessels.
- Pulmonary function tests show non-specific changes such as mild restrictive disease and reduced DLCO, but are important in excluding COPD or restrictive disease as a cause of pulmonary hypertension.
- PA pressures are raised on right heart catheterization, and may be suprasystemic. Pulmonary artery $O_2$ sats are low.
- VQ scans are the single best screening test, but underestimate defects.
- CT and MRI scans are increasingly used.
- Pulmonary angiography is useful but needs expert interpretation.
- Coronary angiography is performed in patients over 45, or with risk factors for coronary artery disease.

## Outcome

Patients remain asymptomatic for months to years. Untreated survival is low and correlates with the degree of pulmonary hypertension:

- 5-year survival for patients with PAP 40mmHg is 30%.
- 5-year survival for patients with PAP >50mmHg is 10%.
- Operative mortality is 5–25%.
- 6-year postoperative survival is ~75%.

## Pulmonary thromboendarterectomy

### *Indications for surgery*

The goals of surgery are prognostic and symptomatic: prevention of RV, prevention of extension of thrombus.

- Symptomatic patients with hemodynamic or ventilatory impairment.
- Marked pulmonary hypertension on exercise.

### *Contraindications to surgery*

- Subsegmental thrombi.
- Severe underlying COPD or restrictive lung disease.
- Advanced age, frailty, comorbidity.

### *Technique of surgery*

The set-up is as described for pulmonary embolectomy (📖 p478), but DHCA is required for complete endarterectomy because of copious flow from bronchial collaterals which develop in chronic pulmonary hypertension. Any lung dissection is technically challenging because of the collaterals from bronchial, diaphragmatic, intercostal, and pleural vessels. A median sternotomy gives access to both sides. Bronchial collateral flow means that venting the left heart is mandatory during cooling.

- During cooling the SVC and right PA are mobilized, and slings placed around the SVC, PA, and aorta.
- The right PA is opened longitudinally, medial to the SVC and any loose thrombus removed, but there is frequently no loose thrombus.
- It is possible to begin the thromboendarterectomy while still cooling, if the bronchial circulation is not copious.
- A microtome knife is used to develop the plane of the endarterectomy, which should be easy if the correct plane is entered, and reveal a pearly white vessel media with no residual plaque.
- If the plane is too deep the reddish adventitia becomes visible, and there is a real risk of perforating the pulmonary artery.
- Each endarterectomy is followed to a segmental branch and should taper off gradually, becoming free spontaneously without breaking off.
- It is possible to resume bypass while repairing the right PA and beginning the dissection on the left.
- ASDs are repaired and other concomitant procedures, including insertion of a Greenfield IVC filter, are carried out during rewarming.

### *Postoperative management*

Moderate to severe RV dysfunction is managed with inotropes and inhaled pulmonary inodilators, e.g., nitric oxide. See also 📖 p191.

# Permanent pacemaker

## Key facts

- Indications for permanent pacemaker insertion (📖 p263) include complete heart block, Mobitz type II and symptomatic Mobitz type I second-degree heart block, as well as trifascicular block.
- Permanent pacemaker insertion may be carried out using a combination of percutaneous Seldinger technique to place endocardial leads via the subclavian veins, as well as open dissection of a pocket usually in the infraclavicular fossa to place the pacing box.
- Occasionally lack of access due to SVC obstruction or occlusion means that placement of endocardial wires is impossible and permanent epicardial leads are placed via a VATs or minithoracotomy approach.

## Placing a permanent epicardial system

***Set-up***

- Ensure that you have the correct pacing system in the OR, and that a cardiac technician is present to test the system as soon as it is in place.
- General anesthesia, with arterial and central line.
- Ensure the patient's chest and abdomen are shaved.
- Prep and drape the patient so that they are exposed from above the umbilicus to the suprasternal notch, as the incision may rarely have to be extended into a median sternotomy to control bleeding.

***Procedure***

- Make a 5–6 cm longitudinal incision running from 1cm above the xiphoid process towards the umbilicus (Fig. 10.1c).
- Cautery down to the xiphisternum, controlling the vein that runs across the sternum at the level of the sterno-xiphisternal junction.
- Continue down the length of the incision as far as the rectus muscle.
- Using a finger, develop a pocket in the plane between rectus and the rectus sheath lying below it, large enough for the pacing box.
- Place a self-retainer in the incision and divide the xiphisternum with Mayo scissors.
- Staying close to the underside of the xiphisternum, use a finger to develop the plane between the sternum and the pericardium: it should be possible to easily feel a ventricular impulse.
- Open the pericardium with a Mackindo scissors, extending the opening as far as required to visualize the RA and the RV, and place stay sutures which can be locked around the retractor.
- If atrial wires are required place these first by retracing the RV leftwards gently using the handles of the Mackindo scissors to expose the base of the RA.
- The easiest pacing system to insert comes with a long-handled applicator and is inserted with a simple punch and twist action.
- Ventricular wires are placed in the same manner, taking care to avoid the PDA and great cardiac vein.

- Take care to identify which wire is atrial and which is ventricular before letting the cardiac technician test them, and then connecting them to the pacing box.
- Place the pacing box in the pocket beneath rectus, tucking redundant lengths of wire into the pericardium.
- Place a redivac drain the pericardium if there is any bleeding and close the soft tissues in layers with 1/0 Vicryl continuous sutures.

# Surgery for atrial fibrillation

## Key facts

- AF increases morbidity and mortality from thromboembolic events (TEs), particularly stroke (📖 p219) despite optimal anticoagulation and rate control.
- Ablation procedures, performed with a variety of energy sources, at the time of cardiac surgery (usually mitral) can restore sinus rhythm in up to 70% of patients at 1 year.
- The Cox–maze cut-and-sew procedure has freedom from AF at 1 year reported to be as high as 98%, and in a few centers is performed as a stand alone procedure for refractory AF.
- LA appendage ligation is a useful adjunct to reduce risk of TEs.

## Pathology

AF is characterized by irregular, disorganized atrial impulses at a rate of 350–500/min, and ineffective atrial contraction. P waves are absent from the EKG. The resulting arterial waveform is irregular in rate and amplitude. Atrial flutter is characterized by regular atrial impulses at a rate of 250–350/min, variable AV block. P waves often have a saw-toothed appearance and the arterial waveform is regular in rate and amplitude. AF has a prevalence of 6% in the over 65s.

- AF is ***induced*** by focal areas of enhanced automaticity in the atria, mostly in and around the *pulmonary veins*, and much less frequently around the SVC and coronary sinus.
- AF is ***maintained by macro re-entry circuits*** throughout the atria which become more persistent the longer AF lasts.
- Treating paroxysmal AF must stop induction pathways, whereas treating permanent AF must address maintenance pathways.
- Ineffective atrial contraction not only reduces cardiac output by up to 30%, but also results in blood stasis which predisposes to mural thrombus, particularly in the ***LA appendage*** (which has a narrower neck compared to the RA appendage).
- Risk factors for AF are age, dilated LA, recent cardiac surgery, recent pneumonectomy, sepsis particularly pneumonia, hypokalemia, hyopomagnesemia, withdrawal of antiarrhythmics.

### AHA classification of atrial fibrillation

- *Recurrent*: two or more episodes.
- *Paroxysmal*: recurrent AF >7 days, terminating spontaneously.
- *Persistent*: recurrent AF >7 days terminated medically.
- *Permanent*: AF that cannot be permanently terminated medically.

## Surgical options

### *Maze procedure*

This is the definitive surgical treatment for chronic AF: it is indicated as a stand-alone procedure in patients that have failed maximal medical therapy and are either symptomatically compromised by AF or have had embolic events. The

***Cox–maze III*** interrupts conduction routes of the most common re-entrant circuits, providing a single pathway from the sinus node to the AV node.

- The name 'maze' arises because conduction to the rest of the atria is maintained by providing multiple blind pathways off the main route, preserving atrial contraction, but preventing re-entrant circuits.
- Conduction is interrupted by creating *full thickness lesions* in the atrial wall: this can be done using incisions which are then resutured, or energy sources including *cryoablation* (employing nitrogen-cooled probes to a temperature of −150°C), unipolar or bipolar *radiofrequency*, or *microwave ablation*.
- The advantage of using *cut-and-sew* is that large atria can be reduced in size, minimizing the risk of macro re-entry circuits. This takes longer than using ablation energy sources (40min to 1h, compared to 10–20min), and the risk of bleeding is greater.
- The advantage of ablation is smaller surgical incisions: unipolar is used in small, delicate areas; bipolar is used over long tracts.
- Set-up requires bicaval cannulation: electrophysiological mapping is not routinely performed preoperatively, as macro re-entry circuits change constantly, and the incisions in the maze procedure (Fig. 11.2) are fixed.

***Modified maze and pulmonary vein isolation***

These more limited lesion sets take less time than performing the classic Maze, and in selected cases can be done without bypass with concomitant OPCAB. Patients have a 60–75% freedom from AF postoperatively, compared to >90% with the cut-and-sew Maze. The disadvantage of pulmonary vein isolation is the risk of refractory atrial flutter in some patients, which may be more symptomatic than AF.

***Biatrial cryomaze at the time of mitral surgery***

This is indicated for patients undergoing mitral surgery, with any previous history of AF. The LA lesion set is performed under cardioplegic arrest, before oversewing the LA appendage (4/0 running Prolene double layer) and placing annuloplasty sutures. The right atrial lesion set can be performed with the clamp on or off. Patients should be started on amiodarone and anticoagulated for 3 months as atrial transport remains abnormal, and the endocardial lesions are thrombogenic.

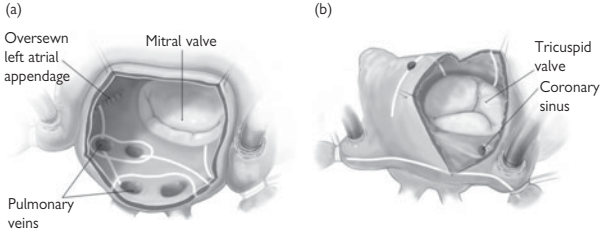



**Fig. 11.2** Pattern of incisions in the Cox–maze III procedure. (a) Left atrium. (b) Right atrium. Reproduced from Chikwe J *et al.* Current concepts in ablation of atrial fibrillation. *Semin Cardiothoracic Vasc Anesth* 2009; **13**: 215–24. © SAGE Publications Ltd./SAGE Publications, Inc., All rights reserved. 2009.

# Surgery for HOCM

## Key facts

- ***Hypertrophic obstructive cardiomyopathy (HOCM)*** is a subtype of hypertrophic cardiomyopathy causing LVOT obstruction, which may be complicated by SAM and mitral regurgitation (📖 p190).
- This can be treated by septal myomectomy ± mitral repair.

## Pathology

Hypertrophic cardiomyopathy (HCM) is a group of heterogenous sarcomere disorders caused by mutations in the β-myosin heavy chain gene on chromosome 14 (q11–q12), in cardiac troponin-T on chromosome 1, and in α-tropomyosin on chromosome 15, transmitted in an autosomal dominant fashion or occurring spontaneously.

- HOCM is characterized by hypertrophy of the LV and the septum, usually maximal in the outflow portion of the ventricular septum, with maximal thickening lying below the free edge of the anterior mitral leaflet (***basal***) or it may be more ***diffuse***.
- Mitral valve leaflets are disproportionately long and thick, and 20% of HOCM patients have MR, usually as a result of systolic anterior motion (***SAM***) of the anterior mitral leaflet, which exacerbates the LVOT gradient which may be as much as 120mmHg.
- The left ventricular cavity is small, with an S shape in young patients.
- Although coronary arteries are usually larger than normal, inadequate capillary flow occurs during diastole because of abnormal relaxation, and septals may be obliterated during systole: myocardial infarction can occur in HOCM in the absence of atherosclerotic disease.
- Although cardiac output is often supranormal in the early stages of disease, progressive fibrosis leads systolic and diastolic dysfunction, compounded by arrhythmias and ischemia.

## Presentation and natural history

Onset of symptoms occurs in any decade and includes angina, effort dyspnea, syncope, and exertional presyncope. LVF results in orthopnea and paroxysmal nocturnal dyspnea. The triad of signs is: (1) bifid arterial pulse, (2) palpable left atrial contraction, and (3) late ejection systolic murmur at the left sternal edge. Although the clinical course may be stable over long periods, sudden death, heart failure, and peripheral embolization may occur. Annual mortality of symptomatic HOCM is about 3%. Risk factors for sudden death include young age, syncope, myocardial ischemia, and sustained VT on ambulatory monitoring.

## Investigation

- EKG shows an LV strain pattern: Q waves, signs of LV hypertrophy, right and left bundle branch block, and atrial hypertrophy also feature.
- CXR shows moderate cardiomegaly, and signs of heart failure including frank pulmonary edema may be present.
- Echocardiography is diagnostic: asymmetric septal hypertrophy and SAM give the diagnosis; and the exact morphology, degree of LV dysfunction, LVOT obstruction and amount of MR can be quantified.
- Coronary angiography is performed to diagnose CAD, and ventricular pressure gradients may be measured directly if TEE equivocal.

## Septal myectomy

### *Indications*

Septal myectomy should be considered in any patient symptomatic despite maximal medical therapy, and whose LV gradient at rest, on exertion, or after an ectopic beat is >50mmHg.

### *Technique of surgery*

Standard bicaval CPB (in case the mitral valve requires repair) under cardioplegic arrest, via a median sternotomy, with monitoring and TEE. The LV is approached via a transverse or oblique aortotomy (Fig. 11.3)

- If the right aortic cusp is carefully retracted, the bulge of hypertrophied septum can be seen and incised in a longitudinal direction exactly below the lowest point of the right aortic cusp. (Fig. 11.3).
- A second incision is made parallel to this as far leftwards as possible, and both incisions are continued down to the apex as far as possible
- A transverse incision is made between the 2 vertical incisions and continued downwards so that a long, 1–1.5cm thick rectangular strip of septum is excised (Fig. 11.3). Angle the blade towards the horizontal not vertical plane to excise maximal muscle. Residual hypertrophy can be digitally palpated and resected using rongeurs.
- *Anomalous papillary muscle*s and thickened chordae which insert into the anterior mitral leaflet should be resected, taking care to preserve primary chordae to avoid causing mitral prolapse and MR.
- Avoid creating a ***VSD*** by not resecting too far towards the RCC/NCC junction where the septum is thinnest, or close to the annulus.
- Once the aortotomy is closed and the heart is de-aired, the gradient across the LVOT can be assessed by TEE: significant gradient warrants further resection, and moderate or worse MR should be corrected.

### *Results of surgery*

Myectomy has a hospital mortality of 2–5%, a 5-year survival of 80–90%, and 10-year survival of 70–85%. Symptom relief depends on relief of LVOTO: 90% of patients are NYHA class I/II postoperatively.

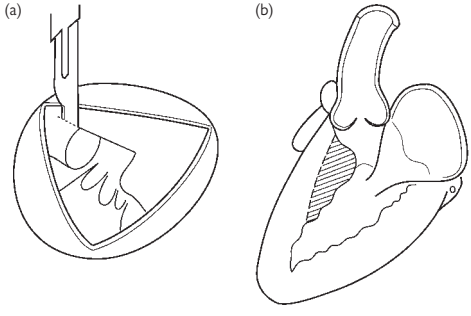



**Fig. 11.3** Technique of myectomy. (a) View through the aortic valve. (b) Cross-section of the LVOT showing the position of the myocardium that is removed.

# Chronic constrictive pericarditis

## Key facts

- Constrictive pericarditis may be idiopathic (the most common) or associated with other conditions including prior cardiotomy (📖 p82).
- Cardiac catheterization shows equalization of LVEDP and RVEDP, a dip plateau pattern in the ventricular filling pressure curve, and a prominent 'y' descent in the JVP, and is needed to differentiate between this and restrictive cardiomyopathy (Table 2.4).

## Pathology

Constrictive pericarditis may be idiopathic or associated with inflammatory or infective processes, or due to connective tissue disorders, systemic diseases including uremia and hypothyroidism, neoplastic, traumatic (including surgery), and physical agents such as radiotherapy. The visceral pericardium becomes a thick rind-like layer, densely adherent to the heart, and preventing normal function resulting in a tamponade-like picture. Occasionally calcifies.

## Presentation and natural history

Constrictive pericarditis may be asymptomatic for years. Fatigue progresses to effort dyspnea, hepatomegaly and ascites with peripheral edema. Orthopnea and paroxysmal nocturnal dyspnea is uncommon. Signs include ↑JVP, ***Kussmaul's sign*** (failure of JVP to fall during inspiration), pulsus paradoxus, systolic retraction in the left parasternal region, and 'pericardial knock'. Salt and water retention is a key feature. The average life expectancy once symptoms appear is 5–15 years.

## Investigation

- Biochemical abnormalities include hypoproteinemia secondary to protein-losing enteropathy, with ↓albumin and gamma globulin.
- EKG shows non-specific ST-segment and T wave changes, and the QRS complexes are low voltage.
- Calcification is present in 40% on the CXR, with enlarged cardiac silhouette in 30%.
- Echocardiography reveals that with inspiration the RV fills supranormally but the LV underfills because of left deviation of the septum, and will also detect thickened pericardium and pericardial effusion.
- Cardiac cath shows elevated RVEDP, equalization between RVEDP and LVEDP, a dip plateau pattern in the ventricular filling pressure curve ('square root sign'), and a prominent 'y' descent in the JVP (Fig. 2.16).
- Constrictive pericarditis can be differentiated from inoperable restrictive cardiomyopathy by a rapid bolus of 1000mL fluid during catheterization which results in: (1) pronounced increase in filling pressures, (2) the typical square root sign, (3) equalization of diastolic pressures, and (4) loss or reversal of the respiratory variation in RAP.
- Endomyocardial biopsy excludes amyloidosis.
- The operative mortality is 2–3%.
- 5-year operated survival is about 65%.
- 5% of patients continue in heart failure after surgery.

## Indications for surgery

The diagnosis of symptomatic constrictive pericarditis is indication for operation. The operative mortality and failure to achieve symptomatic improvement is high in patients with a history of chest irradiation and NYHA status III–IV.

## Technique of surgery

Pericardiectomy can be carried out via a left anterolateral thoractomy, which gives the best access to the LV allowing all the pericardium covering to be removed. Or via a median sternotomy which facilitates access to the RV and CPB. It is possible to carry out pericardiectomy without CPB, although not all hearts tolerate the manipulation necessary to clear the LV without CPB. Pericardiectomy via a median sternotomy is described.

- External defibrillator paddles should be placed.
- The groin should be prepped and draped.
- The pericardium is incised vertically with a 15 blade starting over the RV near the diaphragm where it is easier to find a plane.
- Dissection continues over the ventricular surface, taking care not enter myocardium.
- The right and left phrenic nerve are preserved by dissecting them from the pericardium with a pedicle of fat and soft tissue (Fig. 11.4)
- The plane is extended across the lateral walls of the RA, across entire LV until that is freed as far as the AV groove and diaphragm.
- The cavae do not need to be freed, but the pulmonary trunk does.
- Islands of plaque may be left where pericardium is densely adherent and coronary vessels are likely to be present, such as the AV groove.
- Where it is impossible to separate pericardium from the epicardium, without revealing the dark red, raw bulging myocardium, the pericardium can be cross-hatched to reduce constriction (Fig. 11.4).

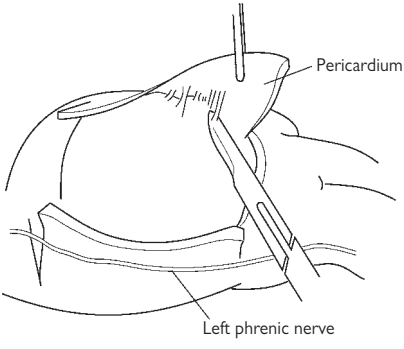



**Fig. 11.4** Technique of pericardiectomy.

# Chronic pericardial effusion

## Key facts

- Pericardial effusions are most commonly due to malignancy, or post-cardiotomy Dresslers' syndrome (📖 p264).
- They can be managed by needle pericardiocentesis: surgical pericardial window is useful for recurrent effusions and for obtaining histology.

## Pathology

The causes of chronic pericardial effusion include:

- Malignancy involving the pericardium (most commonly from breast and bronchial carcinoma), and malignant syndromes.
- End-stage renal failure, dialysis.
- Collagen vascular diseases.
- Dresslers' post-cardiotomy syndrome.

## Presentation

Symptoms may include pericarditic chest pain, angina pectoris, dyspnea, fever, palpitations. Chronic pericardial effusions may be asymptomatic. Signs may include those of cardiac tamponade i.e. ↑JVP, ↓BP, tachycardia, pulsus paradoxus and Kussmaul's sign, as well as a pericardial rub and pedal edema. Chronic pericardial effusion may cause death from tamponade, resolve after pericardiocentesis, or evolve into restrictive chronic pericarditis. Prognosis depends upon the etiology.

## Investigation

- EKG shows widespread concave upwards ST segment elevation if acute pericarditis is present, but in chronic pericardial effusion the EKG is often normal or shows low voltages.
- CXR may show an enlarged, 'globular' cardiac silhouette.
- Echocardiography is diagnostic.

## Indications for operation

- Relief or prevention of tamponade.
- Symptomatic relief.

## Techniques of surgery

***Pericardiocentesis***

This percutaneous approach to aspiration of a pericardial effusion is described on 📖 p246. It is most commonly carried out by the cardiologists. TTE or TEE guidance is a useful adjunct.

***Subxiphoid pericardial window***

3–5cm incision in midline starting over xiphoid and extending inferiorly. Divide upper 1–2cm of linea alba and resect xiphoid with bone nibblers. Smooth any sharp bony fragments with bone file. Retract sternum anteriorly, divide pericardial fat, and identify pericardium superior to diaphragm. Carefully grasp pericardium and nick it, and make a 3×3cm window either laterally so that fluid drains into the pleura, or inferiorly so that fluid drains into the pericardium. A Redivac drain should be placed: this can bed removed after 2–3 days if 24 drainage is <25mL.

***Video-assisted thorascopic pericardial window***

This can be done via a left- or right-sided approach: the advantage of a left-sided approach is that less respiratory compromise is created by dropping the smaller left lung, and the pericardium is closer to the chest wall (helpful if adhesions mean this has to be converted to a minithoracotomy). TEE assessment of where the effusion is largest can decide approach. The approach for left VATS is as described on 📖 p686. The 3×3cm window is created in the pericardium several centimeters anterior to the phrenic nerve, using cautery scissors. If the pericardium does not 'tent' when you grasp it, pick another site to incise as it may be adhesed to the heart. Two intercostal drains are placed via the port sites, and this is placed on suction, and removed when drainage is <50mL/24h.

***Advantages and disadvantages of VATs vs. subxiphoid approach***

- If patient has evidence of ***tamponade***, subxiphoid approach allows quickest relief: if tamponade is marked (hypotension, tachycardia, or RV diastolic collapse on echo) then *surgeon should scrub, prep and drape patient, before induction*, as this can cause loss of cardiac output which will only be relieved by evacuation of effusion.
- Both approaches can result in injury to heart if the pericardium is adhesed at the site of resection: this is easier to control via a subxiphoid approach.
- VATs approach allows aspiration of any pleural effusion, cytology and histology—potentially helpful in malignant disease.
- VATs approach requires set-up for single lung ventilation: in patients with hemodynamic compromise this may not be tolerated.

## Outcomes

- Usually possible to remove drains and discharge patients within 48–72h.
- Perhaps 5–10% recurrence rate depending on technique, size of window and etiology: pleural effusion is usually not problematic.

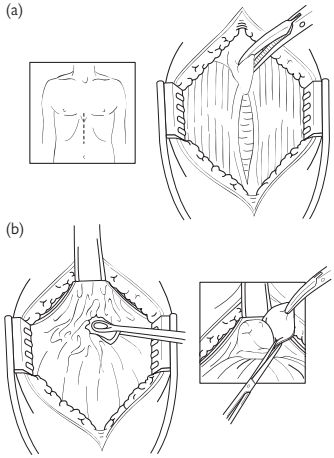



**Fig. 11.5** Technique of subxiphoid pericardial window.

# Heart failure

## Key facts

- Heart failure affects 1–3% of the population, affecting 6 million in the US alone, and accounts for up to 2% of health expenditure.
- The 1-year mortality of NYHA grade IV heart failure is about 60%.
- Ischemic heart disease accounts for around 60% of heart failure.
- Heart failure may be predominantly due to ***systolic*** and/or ***diastolic*** dysfunction, and classed clinically as ***acute or chronic***, ***left and/or right***, and graded by one of three main classification systems (see Box 11.1)

## Etiology and pathophysiology

Pathophysiology is described on 📖 p76. Etiology is detailed on 📖 p76, and varies widely between patient populations, e.g., coronary artery disease accounts for 75% of cases of chronic heart failure in white male patients, but only 30% of cases in females. Dilated cardiomyopathy accounts for around 30% of patients on transplant waiting list, and end-stage valvular heart disease a further 5–10%.

### Box 11.1 Definition of heart failure

The AHA describes heart failure as 'a clinical syndrome that is characterized by specific symptoms (dyspnea and fatigue) and signs (fluid retention. There is no diagnostic test…because it is largely a clinical diagnosis that is based on a careful history and physical examination)'. Many surgical studies use a definition of LV ejection fraction <30%. Diastolic heart failure is defined as symptoms or signs of heart failure, and abnormal LV relaxation, filling or diastolic distensibility.

## Classification of heart failure

The NYHA grading (📖 p2) provides a functional classification which describes symptom severity, is useful for assessing response to treatment, and is widely used in cardiac surgery risk stratification. It was not designed specifically with heart failure in mind and does not correlate well with formal exercise testing, so the ACC/AHA introduced an alternative staging:

*ACC/AHA staging system*

- Stage A: patient at high risk for developing heart failure in the future.
- Stage B: a structural heart disorder but no symptoms at any stage.
- Stage C: previous or current symptoms of heart failure in the context of an underlying structural heart problem managed medically.
- Stage D: advanced disease requiring hospital-based support, a heart transplant, or palliative care.

*UNOS waiting list priority*

In patients listed for heart transplant the UNOS status is used to prioritize management (📖 p507):

- Status 1A: patient on mechanical support.
- Status 1B: patient on inotropic support either in hospital or at home.
- Status 2: patient not on inotropic or mechanical support.

## Presentation

- ***Left heart failure*** is characterized by orthopnea, paroxysmal nocturnal dyspnea, and dyspnea at rest, because of pulmonary edema.
- ***Right heart failure*** presents with hepatic congestion (abdominal discomfort), ascites, and peripheral edema. Congestive heart failure presents with both sets of symptoms.
- Many patients may present immediately or months to years following an MI, with exertional fatigue, dyspnea, and fluid retention.

## Investigation

- EKG may show Q waves from old infarctions, AF, wide QRS.
- CXR may show cardiomegaly, pulmonary edema, pleural effusions.
- Echo is diagnostic and quantifies LV and RV function and dimensions, regional wall motion abnormalities, dilatation, hypertrophy, and valvular disease. Exercise or dobutamine stress echo can further identify contractile reserve and hibernating myocardium.
- Viability can be assessed using thallium or technetium (📖 p28).
- MRI is useful to quantify ventricular and valve function and dimensions, ischemia and infarcts (using gadolinium) and viability (dobutamine).
- Deranged LFTs, electrolyte (↓$Na^+$, $K^+$) and hematological (↓Hb) abnormalities are not unusual.
- Right heart catheterization is indicated to assess PVR, PAP, and cardiac output, and coronary angiography is performed in at-risk patients to identify coronary disease.

## Medical management

Treatment is directed at treating risk factors and underlying causes, symptom relief, and improving prognosis.

- ***Loop diuretics*** (furosemide 40–120mg daily, or bumetanide 1–4mg daily) are titrated to response to relieve symptoms of fluid retention, together with potassium-sparing diuretics (spironolactone 50–200mg daily, or amiloride 5–10mg daily) to prevent hypokalemia, and thiazide diuretics (metolazone 2.5–5mg once or twice weekly) in resistant cases.
- ***Nitrates*** (ISMN 30–120mg daily) improve symptoms by a minor reduction on systemic and pulmonary vascular resistance, and improved coronary perfusion.
- ***ACE inhibitors*** (captopril 12.5–50mg three times daily, enalapril 10–20mg twice daily) improve symptoms and prognosis through afterload reduction, but patients with renal dysfunction or on potassium-sparing diuretics must be monitored closely for hyperkalemia.
- Careful use of β-***blockers*** (bisoprolol 1.25–5mg twice daily, carvedilol 3.125–10mg twice daily) improves symptoms and long-term prognosis through a favourable effect on diastolic function.
- ***Digoxin*** (62.5–375 micrograms daily) has a mild positive inotropic effect in addition to its antiarrhythmic effect.
- ***Atrio-biventricular pacing***, which restores synchronous ventricular contraction, may improve symptoms in patients with widened QRS, and CHF patients should be considered for prophylactic *AICD*.

# Surgery for heart failure

## Key facts

Surgical options for heart failure include revascularization, valve surgery, LV remodeling, VADs (📖 pp498–501), and transplantation (📖 pp506–519).

## Coronary artery bypass grafting

Coronary artery disease (CAD) is the commonest cause of heart failure. Operative mortality for isolated CABG in selected patients with chronic severe LV dysfunction is around 5%, compared to patients with preserved LV function where it is 1–2%, and long-term survival is poorer. However, selected patients with severe LV dysfunction and CAD still derive significantly more benefit from surgical revascularization than they do from PCI or medical management, in terms of symptoms, freedom from revascularization and survival. Patients with poor LV being considered for CABG should have: (1) ***severe, obstructive CAD*** with (2) ***adequate targets***, ***and*** documented evidence of (3) ***ischemia*** and (4) ***substantial viability***. If dyspnea and other failure symptoms predominate over angina then the benefits of CABG are less clear. Patients with severely dilated LV have low likelihood of showing improvement in LVEF, even if they have large areas of viability: concomitant ventricular restoration (📖 p495) may have a role in ameliorating symptoms of heart failure.

***Adjuncts to standard management in CABG patients with poor LV***

- Preoperative assessment of myocardial viability:
  - Technetium and thallium SPECT, or PET (📖 p28).
  - Dobutamine stress echocardiography (📖 p25).
  - MRI (📖 p29) dobutamine or gadolinium.
- Preoperative nesiritide or NTG infusion to reduce RV afterload, improve myocardial perfusion, and treat pulmonary edema.
- Prophylactic IABP may decrease mortality but carries substantial morbidity in arteriopaths.
- Aggressive myocardial protection with blood cardioplegia given antegradely, retrogradely and down grafts at frequent intervals.

## Mitral valve surgery

MR may be the cause or the consequence of heart failure. Organic leaflet disease, e.g., leaflet prolapse causing type II MR (📖 p400) can eventually result in severe LV dysfunction as a result of chronic volume overload. Type IIIb MR (📖 p356) results from LV dilatation which causes posterior displacement of the posteromedial papillary muscle and restricted closure of normal leaflets, as a result of ischemic cardiomyopathy. Dilated cardiomyopathy causes annular dilatation resulting in failure of normal leaflets to coapt causing type I MR. Deciding whether a patient presenting with severe MR and severe LV dysfunction will benefit from surgery or has 'missed the boat' is a common dilemma, particularly as LV function is overestimated in the presence of severe MR:

- Although the operative mortality is 5–15% and patients will still have severe LV dysfunction, surgery is likely to improve symptoms, and may improve survival in patients with LV dysfunction due to MR, if the valve can be repaired. In valves that must be replaced chordal preservation is necessary to achieve prognostic and symptomatic benefit.

- Where MR is secondary to severe LV dysfunction the advantages of mitral surgery over optimal medical management (e.g., β-blockers, ACE inhibitors) and ventricular resynchronization have not been demonstrated. Certain subsets of patients, e.g., those with ischemic cardiomyopathy, suitable targets and evidence of viability; gain symptomatic and prognostic benefit from repair (with a rigid, downsized, remodeling ring).

## Left ventricular remodeling

This includes a number of procedures, all of which are based on ***Laplace's law*** (reduction in LV radial dimensions lead to a reduction in wall tension, and hence improvement in ventricular function).

***The Batista procedure (partial left ventriculectomy)***

Designed for DCM. Largely been abandoned because of high mortality and morbidity. It involves resection of the posterolateral LV wall, between or including papillary muscles. Mitral repair is sometimes performed from the ventricular aspect. DCM does not affect the LV homogenously: improved outcomes have been reported when this procedure is limited to patients in whom the lateral LV wall is worst affected.

***Dor procedure (left ventricular restoration)***

Designed for ischemic cardiomyopathy, the *Dor procedure* excludes scarred dyskinetic and akinetic segments, reshaping the ventricle from its pathological spherical shape and restoring its more physiological elliptical shape. It is usually performed with full revascularization ± mitral repair.

- The LV is opened through scar tissue, subtotal endocardiectomy is performed, the scar tissue is excluded using a circumferential endoventricular purse-string, leaving a smaller LV cavity (often sized using shaped balloon inflated with saline to a volume 50–60 × BSA), and the residual apical defect closed with a Dacron patch (📖 p351).
- Operative mortality has been reported at <2%, with 3-year survival of over 75%, and sustained improvement in NYHA class.
- The STICH trial[1] found that adding SVR to CABG did not improve mortality, symptoms or hospitalization for CHF compared to CABG alone in these patients, but there are several problems with the data (📖 p349).

## Cellular cardiomyoplasty (cell transplantation)

These techniques, which are in the early clinical trials, involve implanting cultured suspensions of cardiac, skeletal, or smooth muscle cells; or progenitor cells harvested from bone marrow into injured myocardium.

- Can be done percutaneously with cells infused into coronary arteries in infarcted territory; or directly during surgical revascularization.
- Initial experience suggests that implanted cells engraft within the infarcted region but evidence of clinical benefit from improved regional perfusion, wall thickening and motion, and global motion is lacking.
- Skeletal myocytes implanted into an infarct region adopt a similar morphology to cardiac myocytes but do not have synchronous contractility, and may contribute to arrhythmias.

## Reference

1. Jones RH *et al.* Coronary bypass surgery with or without surgical ventricular reconstruction. *NEJM* 2009; **360**:1705–17.

# Ventricular assist devices

## Key facts

These are devices that unlike IABP (📖 p192), cardiopulmonary bypass (📖 p102), and ECMO (📖 p500), are anastomosed to the appropriate atrium and outflow vessels to provide augmentation of ventricular outflow.

## Types of ventricular assist devices

*Side of heart assisted: left, right and biventricular*

- The left ventricular assist device (*LVAD*) receives inflow from the LA or LV, and outflows into the aorta.
- The right ventricular assist device (*RVAD*) receives inflow from the RA or RV and outflows into the pulmonary artery.
- The biventricular assist device (*BiVAD*) consists of an RVAD and an LVAD attached in parallel, or a dual chamber device known as a total artificial heart (*TAH*), e.g., Abiomed.

***Pulsatile versus non-pulsatile***

- Pulsatile: most devices employ pneumatic pumps to drive pusher plates or compress polyurethane sacs giving stroke volumes of up to 85mL (e.g. *Thoratec PVAD, IVAD, AB5000, Berlin heart*), but some electric devices, e.g., *Novacor, Heartmate XVE* are also pulsatile.
- Non-pulsatile devices may be axial, centrifugal, or bearingless magnetic levitators and are generally smaller with fewer moving parts and less surface area in contact with blood, and therefore less thrombogenic, less prone to driveline infections, quieter, and more durable than pulsatile devices.
  - Axial flow pumps include *Heartmate II, MicroMed DeBakey, Jarvik 2000, Impella* (small enough to be inserted percutaneously).
  - Centrifugal devices: *Levotronics, Ventrassist, Tandem Heart.*
  - The *Duraheart* is a bearingless magnetic device.

***Intracorporeal (implantable) vs. extracorporeal (external devices)***

- Most devices can be implanted, usually intra- or extraperitoneally.
- VADs using axial flow technology (Jarvik 2000, Thoratec HeartMate II, and the DeBakey VAD) can be implanted into small patients.
- VADs used for short-term support, e.g., Centrimag are usually left externally. Pediatric devices, e.g., Berlin Heart, are usually external.
- Electromagnetic flow coupling removes the need to have a power line passing externally, reducing the risk of infection which is the main cause of complications leading to VAD failure or removal.

## Indications

These devices are currently indicated for:

- *Bridge to recovery, bridge to decision* (for temporary support in patient with cardiogenic shock with questionable neurologic or end-organ status), ***bridge to bridge***:
  - Ultra-short-term, e.g. post AMI cardiogenic shock: Impella, Tandem Heart (both percutaneous, deliver 2–3L cardiac output).
  - Short term, e.g. post cardiotomy cardiogenic shock, myocarditis: Levitontrix Centrimag).

  - Short- to long-term, e.g., acute fulminant myocarditis: Thoratec PVAD, AB 5000.
- *Bridge to transplantation*: Heartmate, Heartware, Duraheart, Ventrassist, Debakey.
- *Destinational* (permanent) therapy for patients with refractory heart failure who are not eligible for transplant: Heartmate, Duraheart, Ventrassist, Debakey.

## Management of cardiac disease in VAD candidates

***Coronary artery disease***

Management of CAD depends on the indication for VAD insertion: in high-risk patients undergoing VAD insertion to support revascularization CABG is carried out siting top-ends on the lesser curve of the aorta, leaving plenty of space for LVAD insertion. For implantable long-term VADs, left-sided grafting is usually not performed, but RCA grafting may be indicated to avoid RV failure post LVAD insertion (📖 p499).

***Valvular heart disease***

- *Aortic incompetence* >2+ must be treated to prevent blood regurgitating back to the device: leaflets are partially oversewn or bioprosthetic valve replacement is performed.
- *Aortic stenosis* does not need treatment in bridge to transplant patients, but in bridge to recovery patients using a bioprosthetic rather than mechanical valve prosthesis reduces risk of thromboembolism.
- *Mitral regurgitation* generally does not need correction.
- *Mitral stenosis* is addressed to avoid a low output syndrome resulting in the reduction in LV inflow: a bioprosthetic replacement.
- *Tricuspid regurgitation* may improve with improvements in right heart failure, but severe TR should be addressed with annuloplasty (📖 p417).
- *PFO* closed if present.
- A *mechanical AVR* is replaced with a bioprosthesis to reduce risk of thromboembolism associated with stasis across the AVR; if there is a *mechanical MVR* this is left *in situ* and the patient anticoagulated.

## Principles of LVAD implantation

- Median sternotomy or sternum-sparing incisions are used: VADs can be implanted on- or off-pump. The pocket for implantable VADs is intra-or extraperitoneal, driveline tunnelled across abdominal wall.
- LV apex is used for inflow: any thrombus is removed and interrupted pledgetted 2/0 Ethibond mattress sutures are placed circumferentially around the ventriculotomy, then through the sewing cuff of the inflow cannula, and tied down.
- The outflow cannula is anastomosed to the greater curvature of the ascending aorta after longitudinal aortotomy with a side-biting cross-clamp. This is done proximally to leave adequate distal aorta for transplantation. The outflow graft is curved gently to the right to avoid kinking and proximity to the midline (facilitating future resternotomy).
- During and after connection of the graft to the device, de-airing is performed though the graft and ascending aorta.
- The VAD is initiated at low speed on partial CPB and full speed is achieved when off CPB (to avoid entraining air).

# The VAD patient

## Basic principles

- In ***LVAD****s* the LV is connected to the ascending aorta via a pump (📖 p496) unloading the LV, optimizing myocardial $O_2$ supply:demand.
- The aortic valve should open only intermittently: pump output equals cardiac output and the arterial trace is flatline in a non-pulsatile device.
- LVADs deliver up to 10L/min, percutaneous devices (e.g., Impella and Tandem) deliver 2–3L/min.
- In ***RVAD****s* the RA is connected to the PA via a pump (📖 p496), which unloads the RV allowing it to recover function. Satisfactory LV function is required for adequate systemic perfusion. RVAD flow is initiated at 2.0–2.5L/min to achieve LAP and RAP 8–16mmHg.
- In ***biventricular assist devices*** *(BiVADs)* the RVAD and LVAD connected as described earlier, operating in series.
- Flow is determined at the console: alterations in heart rate and filling lead to a compensatory change in stroke volume. A cardiac index of 3.0–5.0L/min, LAP and RAP 10–12mmHg are usual targets.
- LVAD and RVAD rely on adequate function of the unassisted ventricle.

## Anticoagulation management for VAD patients

- Postoperative anticoagulation differs between VADS and institutions.
- ***Percutaneous short-term devices*** (e.g., Impella, Tandem Heart) iv heparin is started immediately.
- ***Implantable*** *VADs* (e.g., Jarvik, Thoratec Heartmate II)) usually no anticoagulation is given for first 24–36h immediate postoperatively. Aspirin and warfarin is started after that if no excessive bleeding. Intravenous heparin can be used until INR 2–3.0. Platelet studies can be performed to see the effectiveness of aspirin.
- ***Surgical temporary assist devices*** (e.g., Centrimag) iv heparin (target APTT 70–90) is usually started 24–48h postoperatively (earlier in LVADs/BiVADs as compared to RVADs). Lower APTT (target 50–70) is required for heparin-bound circuits and cannulas.
- A higher APTT is required in low speed operation or weaning mode.

## Common problems

### *Inability to achieve satisfactory flows*

- May be due to hypovolemia, kinking of the inflow or outflow lines, failure of the unassisted ventricle, ventricular arrhythmias, and thrombosis of the device. Check that the external inflow and outflow lines are not kinked. Ensure that the patient is optimally filled and correct arrhythmias. Get a TEE to diagnose tamponade, failure of non-supported ventricle, malplacement of the inflow cannula.

### *Mediastinal bleeding*

- 20–50% of VAD patients bleed excessively postoperatively and reoperation rates for bleeding are high. Multiple factors contribute including surgical suture lines, emergency and reoperative surgery, coagulopathy, or anticoagulation. Antifibrinolytics are given on induction, and meticulous surgical hemostasis is carried out in an

attempt to reduce bleeding complications. If the surgeon is confident there is no surgical source, any coagulopathy is corrected aggressively: continued excessive bleeding requires re-exploration. Massive transfusion is described on 📖 p149.

***Right ventricular failure***

Occurs in 10–20% of patients with LVADs. Suggested by rising CVP, falling VAD flows. RVEDP equal to CVP. TEE diagnostic. It is associated with large volume transfusion (may be cytokine mediated). Medical management—avoiding volume overload. Phosphodiesterase inhibitors, inhaled nitric oxide or prostaglandins, inotropes to increase RB contractility and reduce afterload through lowering PVR. RVAD is required in refractory cases with low RVSWI (RV stroke work index) with higher CVP and lower PA pressures.

***Thromboembolism***

The incidence is about 5–15 %, higher in the setting of clinically important infections. Head CT to rule out hemorrhage before increasing APTT.

***Management of sepsis in VAD patients***

Nosocomial infections result in prolonged hospitalization, poor nutritional status, immune suppression from chronic illness, and multiple indwelling monitoring lines. Prevention is critical: strict asepsis during surgery and any intervention, contact precautions, driveline site care, cultures whenever clinical features of infection and appropriate antibiotics. Sepsis is aggressively treated with antibiotics according to sensitivity. For implantable devices driveline site or pump pocket debridements, explant of VAD or urgent transplantation may be required.

***Other problems***

- *Renal or hepatic dysfunction*: occurs in 1/3 of cases (📖 p236, p276).
- *Hypoxia*: inline oxygenator can be temporarily added to some circuits.
- *Hemolysis*: occurs in ~5 % cases. Patients have anemia, high indirect bilirubin, LDH and free plasma hemoglobin, low haptoglobulin levels. Reduction in VAD speed, and correction of anemia is usually sufficient.
- *Device failure*: usually needs device exchange. Can be explanted.

***Weaning VAD support***

Weaning is performed with echo assistance (watching LV, RV, septum, volume status). CVP,MAP, PAP, PCW, mixed venous sats are observed for normal/acceptable values. Adequate anticoagulation must be ensured. Small inotropic support is usually required.

## Conduct of a code when a patient has a VAD

- ABC: call for help, protect the airway, give $O_2$—remember VAD patients may stroke, aspirate, and have respiratory causes of arrest.
- Try to feel pulse, assess BP, assess VAD function (look at console and any alarm messages, auscultate pump), assess rhythm:
  - If power failed, may be possible to switch to back-up or hand pump.
  - If no adequate BP *start chest compressions*: risk of rupturing heart is lower than immediate risk of no output.
  - Shock VF/VT: no need to disconnect device.
- Get iv access, give epinephrine and volume, get arterial access and assess $PaO_2$, $PaCO_2$, and pH and treat aggressively.

# Extracorporeal membrane oxygenation

## Key facts

- ECMO, also known as extracorporeal life support (ECLS) is based on the same principles as CPB (📖 p192), with several key modifications that allow support to continue for days rather than hours.
- Its primary role is in management of acute respiratory insufficiency in children: it is used as a salvage strategy in adults.
- ECMO is usually instituted via the groin, open or percutaneously, but can also be instituted via central cannulation.

## Basic principles

- There are two main types of ECMO: ***venovenous*** provides respiratory support, and ***venoarterial*** provides both respiratory and hemodynamic support.
- A typical ECMO circuit consists of:
  - Heparin-bonded circuit usually connecting to femoral venous cannula to the pump and oxygenator, with inflow to a femoral arterial or venous cannula depending on whether it is venovenous or venoarterial ECMO.
  - A hollow-fiber oxygenator with integrated heat-exchanger.
  - Centrifugal pump (these are afterload and inflow dependent, i.e., if downstream or inflow obstruction occurs pump flows will decrease).
- An ECMO circuit differs from a CPB circuit in several regards, that means it can be used for longer periods and with lower heparin levels without the same degree of inflammatory response, coagulopathy, or thromboembolic complications.
  - Use of heparin bonding and hollow-fiber membrane oxygenators.
  - The absence of reservoirs, eliminating blood stasis.
  - The absence of cardiotomy suction and air entrainment, reducing hemolysis and activation of inflammatory and clotting cascades.

## Indications for ECMO

### *Adults*

- Severe acute potentially reversible respiratory failure: the CESAR (Conventional ventilatory support versus ECMO for Severe Acute Respiratory failure) trial randomized 180 adults to ECMO or conventional management, with better outcomes at 6 months in the ECMO group. Patients >65 years were excluded.
- No large-scale studies of ECMO for cardiac failure in adults: survivals reported around 20–40% depending on indications for venoarterial ECMO in patients post cardiac arrest, severe cardiogenic shock or failure to wean from CPB.
- It is occasionally used as a bridge to transplant in place of a VAD.

### *Children*

- The indications for ECMO in children are outlined on 📖 pp538–539.

## Technique

The technique used very much depends on the setting (e.g., bedside emergency vs. failure to wean from CPB), and whether venovenous (respiratory support only) is intended, or venoarterial for respiratory and hemodynamic support. Generally institution via peripheral percutaneous cannulation is fastest in the bedside setting, when surgical instruments and assistance is not available. In the operating room, femoral cut-down (especially when peripheral pulses are not palpable due to obesity or low output states) allows rapid, accurate placement of both cannulas.

- The CPB cannulas can be used: they are normally reinforced with additional purse-strings and snares.
- Alternatively femoral cannulation—the patient can be weaned and ECMO discontinued at a later date without reopening the chest.
- Advantages of closing the chest include reduced bleeding and infection risk, less need for sedation and repeat anesthetic, and more rapid weaning from the ventilator if that becomes an option.

***Failure to wean from cardiopulmonary bypass***

- A femoral cut-down is made as for peripheral cannulation (📖 p120). If the patient is stable on CPB an 8mm Dacron graft can be sewn onto the femoral artery, and cannulated with a 14 Sarns: this ensures adequate distal as well as antegrade perfusion.
- Alternatively an 18F Fem Flex II can be inserted through the skin and into the femoral artery under direct vision using a modified Seldinger technique: arterial cannulas inserted in this way should be removed under direct vision, with closure of the arteriotomy with 6/0 Prolene, as the lack of surrounding tissue precludes hemostasis by pressure.
- Axillary cannulation can also be done in the same way, and is useful option in the patient with peripheral vascular or aortic disease
- As large a venous cannula as possible (24–28F) is inserted into the femoral vein using the same modified Seldinger technique.
- The cannulas are carefully de-aired and connected to the circuit, which has also been fully de-aired, bypass is discontinued and the ECMO pump is slowly increased to flows of 4–6L.

***Venovenous ECMO***

- Percutaneous insertion is generally used, one cannula in each femoral vein: the largest possible venous cannula (24–28F) is directed into the RA under TEE guidance.
- A smaller cannula returns oxygenated blood from the ECMO circuit.

## Management

- Oxygenation is dependent on blood flow in venovenous ECMO: near maximum flow rates may be required.
- LV function may worsen: ECMO does not offload the LV as effectively as an LVAD, and progressive low cardiac output may result in LV distension, pulmonary hypertension, and eventually catastrophic pulmonary hemorrhage.
- Inotropes and an IABP may be indicated, along with LV decompression (percutaneous balloon atrial septostomy).

# Basic transplant immunology

Allorecognition is a by-product of a system that evolved to recognize and eliminate infectious agents, by identifying peptide fragments as 'self' or 'non-self'. The mammalian immune system is conventionally considered in two parts: innate and adaptive. Allorecognition is a function mainly of the adaptive immune response.

## Allorecognition

Allorecognition depends on two main processes. Firstly, alloantigen must be taken up and complexed with surface MHC molecules on antigen presenting cells (APCs). Direct recognition is where donor APCs within the allograft perform this function. Indirect recognition is where host APCs do this. Secondly, antigen-specific host T cell receptors must bind to the peptide fragments of alloantigen complexed with the MHC molecules on APCs, which include B cells, dendritic, and Langerhans cells.

## B- and T-cell diversity

APCs are able to identify, take-up, and present alloantigen because of the presence of B-cell receptors (BCRs) specific for each alloantigen. Similarly T-cell receptors (TCRs) are specific to each alloantigen. The production of a huge number of receptors (approximately $2.5 \times 10^7$), each specific to a particular peptide fragment, is possible because of the millions of potential combinations of the thousands of TCR and BCR gene segments. The TCR gene segments are located on chromosomes 14 and 7; the BCR gene fragments are on chromosome 14.

## T-cell activation

When the TCR binds to antigen complexed with MHC molecules a chain of reactions between intracellular signalling molecules leads to cell apoptosis unless CD4 or CD8 also binds to the MHC. There are a number of other co-stimulatory molecules. Binding of the TCR as well as CD and co-stimulatory molecules with the MHC leads to T-cell activation, differentiation and clonal expansion. This results in the rapid production of large numbers of cells able to coordinate and effect destruction of tissues bearing the specific alloantigen: this manifests clinically as rejection.

## Rejection

There are several types of rejection:

- *Hyperacute rejection* (HAR) is mediated by preformed antibodies that bind to antigens of ABO blood groups, non-self HLA, and xenografts similar to antigens found on bacteria and viruses, and results in immediate tissue edema, hemorrhage, and thrombosis.
- *Acute rejection* is a function of both the innate and the adaptive immune system, triggered by the recognition of foreign MHC and foreign peptides presented by self-MHC, by T cells, and results in tissue destruction over days to many months after transplant.
- *Chronic rejection* is a poorly understood vasculopathy which occurs over years: it is probably a function of the adaptive immune system mediated predominantly through indirect recognition.

## A glossary of terms in transplant immunology

***Adaptive immunity:*** learned response to specific non-self antigens.

***Allo-:*** tissue from a genetically different member of the same species. Alloantigen: antigen from a genetically different member of the same species. Allogenicity: ability of tissue to provoke an immune response when transplanted into genetically different member of the same species. Allorecognition: recognition of alloantigen as non-self.

***Antigen:*** cell surface glycoproteins.

***Antibody:*** specific proteins produced by B cells in response to non-self antigen, consisting of two light and two heavy chain proteins, composed in a constant and a variable region. The variable chain region binds with the antigen that triggered the response, and the constant region co-ordinates the cellular response.

***APC (antigen presenting cells):*** these cells ingest and process antigen, then present it bound to surface MHC to T cells. APCs are most commonly found B cells, but other leukocytes can act as APCs.

***B cells:*** B cells mature in bone marrow. They are APCs which can produce antibodies. In response to antigen they undergo clonal expansion, triggered and coordinated by T-helper cells.

***Cellular immunity:*** adaptive immunity.

***CD:*** cellular differentiation molecule (followed by a number, e.g., CD4).

***Clonal expansion:*** production of large numbers of identical cells.

***Co-stimulatory molecules:*** molecule receptors that must be activated, in addition to a main receptor, for a process to happen.

***Direct recognition:*** donor APCs present alloantigen to host T cells.

***Indirect recognition:*** host APCs present alloantigen to host T cells.

***Innate immunity:*** instant inbuilt response to certain non-self proteins.

***HLA (human leukocyte antigen):*** human MHC.

***Humoral immunity:*** innate immunity.

***MHC (major histocompatibility complex):*** glycoproteins expressed on the surface of all cells, unique to that individual, coded for by the MHC genes on chromosome 6. Alloantigen must be bound to MHC for T cells to recognize it. There are two classes of MHC. Class I (HLA molecules A, B, and C) is present on all cell membranes. Class II (HLA molecules DP, DQ, and DR), also known as minor histocompatibility complex, is present on only certain cell types. Class II is less allogeneic.

***Rejection:*** destruction of tissue by host immune response (📖 p502).

***T cells:*** T cells mature in the thymus. T cells that bind to thymic tissue (self) are destroyed. T-helper cells, which are CD8 +ve, coordinate and T cytotoxic cells (CD4 +ve) effect the response.

# Patient selection

## Key facts

The following guidelines for patient and donor selection are taken from the International Society of Heart and Lung Transplantation (ISHLT) Recommendations for Donor Heart Selection 2010.

### Indications for heart transplantation

- Refractory heart failure (stage D).
- Ischemic heart disease with intractable angina not amenable to revascularization or responsive to maximal medical therapy.
- Intractable arrhythmia not amenable to ablative therapy, not responsive to maximal medical therapy.
- Hypertrophic cardiomyopathy that has failed maximal medical and surgical therapy.
- Congenital heart disease.
- Cardiac tumor confined to myocardium without metastases.

### Contraindications to heart transplantation

- Age >70 years.
- Fixed pulmonary hypertension.
- Life threatening, life-limiting illness, e.g., HIV, malignancy.
- Sepsis including endocarditis.
- Active peptic ulcer.
- Continued abuse of alcohol or other drugs.
- Irreversible secondary organ failure not appropriate for combined transplant.
- Diabetes with end-organ dysfunction.
- Multisystem SLE/sarcoid.
- Psychiatric history likely to result in non-compliance.
- Severe peripheral or cerebrovascular disease.
- Malignancy (except cardiac with low probability of metastasis).
- Lack of psychosocial support.

### Patient assessment

Once a patient has been referred assessment normally takes place in two main stages: an initial detailed outpatient assessment designed to avoid invasive and costly investigations in a patient who is clearly unsuitable for transplantation, followed in appropriate cases by a 2–4-day inpatient assessment. The goals of the inpatient assessment are:

- To assess the patient's clinical, psychological, and social suitability.
- To ensure that the patient and relatives are fully informed about all aspects of being on the transplant list, the procedure, and aftercare.
- To familiarize the patient with the transplant unit and staff.
- To optimize the patient's clinical, psychological, and social condition.

## Accepting a patient on to the transplant list

The results of these investigations are considered at a multidisciplinary meeting attended by cardiothoracic surgeons, cardiac anesthesiologists, cardiologists, respiratory physicians, transplant coordinators and nurses, dieticians, and physiotherapists. This is usually after the patient's discharge. The patient is informed of the decision and receives:

- A detailed explanation of the waiting list procedures (including their responsibility to be available for potential transplant at all times, and duty to inform the transplant team of any changes in their health, circumstance, medication, and planned holidays).
- A booklet describing this in more detail, and explaining what to do when called for surgery, the operation, accommodation for partners, publicity and the media, wards, and departments after surgery.

### Routine investigations in transplant assessment

- A detailed history and physical examination.
- CXR, lung function testing, chest CT if >65 years.
- Cardiovascular evaluation:
  - 12-lead EKG, Holter, echo.
  - Thallium, PET or MRI for viability.
  - Right and left heart catheterization.
  - Exercise stress with $VO_2$max.
  - Myocardial biopsy for CHF of unknown etiology.
- FBC, ESR, APTT, PT, INR, fibrinogen, reticulocytes.
- Plasma viscosity.
- U&Es, creatinine, creatinine clearance, urate, calcium, phosphate, liver enzymes, cardiac enzymes, fasting blood glucose and lipids, amylase.
- Screening for malignancies: PSA, mammogram, PAP smear, endoscopy as indicated.
- MSU, urinalysis, nose swab and MRSA screen.
- 24h urine, creatinine clearance, renal function, urinalysis, ultrasound and pyelogram as indicated.
- Full dental assessment and repair, bone densitometry.
- Serology for Hep B/C, HIV, syphilis, rubella, EBV, herpes varicella and zoster, CMV, toxoplasma, fungal, parasites.
- AMF, anti DNA, SCAT/LATEX.
- HLA typing, lymphotoxic and blood group antibody screen.
- Neuropsychiatric assessment.
- Physiotherapy and social work assessment.

# Donor selection and matching

## Key facts

- Donor hearts are ideally from patients aged <45 years, with LVEF >40% on minimal inotropic support, with absence of structural or functional valve disease (including endocarditis) and no coronary disease.
- The alternate list provides less optimal donor hearts: from patients >55 years of age, with valve disease that can be treated by bench repair or replacement, or coronary disease that could be bypassed.
- An estimated ischemic time of <4h is preferable.
- The donor body weight should not be <30% of the recipient's weight in females (20% in males), and hearts must be ABO compatible.

## Donor selection

Patients are considered for organ donation if there is evidence of brainstem death (apnea, absent papillary reflexes, irreversibility over 24h, and absence of reversible causes) and prior consent. Potential donors undergo a review of their history and clinical examination, EKG, CXR, ABGs, ABO typing, testing for HIV, hepatitis C and B, HCV and CMV, and echocardiography. The shortage of donor organs means the criteria for donor suitability are elastic:

- ***Age <45*** ideal, 45–55 years acceptable. >55 years in specific situations.
- ***LVEF >40%***, no major regional wall motion abnormalities, no intractable arrhythmias, no excessive inotropes. Absence of EKG findings of LVH, wall thickness<14mm on echo.
- ***Valves***: absence of structural or functional valve disease (bench repair most commonly of tricuspid valve and replacement acceptable).
- ***Coronaries***: absence of obstructive disease of major coronary arteries (concomitant CABG can be considered for alternate list).
- ***Estimated ischemic time***: <4h preferable, but can be more when other conditions favourable, and in sparsely populated areas, e.g., in Canada longer ischemic times are the norm.
- ***Toxins:*** cocaine use OK if cardiac function acceptable, and no LVH. History of alcohol abuse usually not acceptable. Poisoning acceptable if all other parameters OK.
- ***Sepsis:*** acceptable if recent cultures negative after treatment with antibiotics according to sensitivities, and no evidence of endocarditis. HIV is an absolute contraindication; Hep B and C are relative contraindications.
- Absence of prolonged cardiac arrest, prolonged ↓↓BP, pre-existing cardiac disease, chest trauma, or intracardiac drug injection, extracerebral malignancy.

## Matching donors and recipients

The shortage of donor organs led to the formation of authorities designed to coordinate the transplant recipient list and donor availability, so that organs are allocated to the most suitable recipient, e.g., UNOS (United Network for Organ Sharing) in the US, UK Transplant in the UK. The following factors are taken into account:

## Logistics

Geographical limitations are important to minimize ischemic time to <4h wherever possible.

## Priority

In patients listed for heart transplant the ***UNOS status*** is used to prioritize management:

- ***Status 1A:*** patient admitted in listing transplant centre with at least one of the following:
  - VAD ≥30 days, or TAH, IABP or ECMO.
  - Mechanical circulatory support <30 days, but device-related complications.
  - Mechanical ventilation.
  - High-dose or multiple inotropes, or continuous hemodynamic monitoring.
  - Life expectancy <7 days.
- ***Status 1B:*** VAD <30 days without complications, or continuous iv inotropes.
- ***Status 2:*** all other actively listed patients.

## Histocompatibility

### *ABO compatibility*

Donor and recipient must be ABO compatible: hyperacute rejection (HAR) occurs in ABO incompatible patients. Flow cytometry ABO testing is more sensitive than cytotoxic testing. In infants, with underdeveloped complement system and low levels of isohemagglutinins, ABO mismatched transplants have been carried out without *HAR* and with good long-term outcomes. Some have extended use to children 2 years of age.

### *HLA typing*

Although hearts are amongst the least allogeneic organs, and a HLA mismatch is not a contraindication to transplantation, HLA-$A_2$ or $A_3$ mismatch has been associated with chronic rejection, and some centers choose to avoid this. HLA typing is important as it guides immunosuppression postoperatively. Pre-formed antibodies in the recipient sera are detected by adding patient sera to a panel of 40–60 HLA antigens. This is called a *lymphocyte toxicity screen*. If more than 10% show reactivity a lymphocyte cross-match between recipient candidate to donor must be performed, as pre-formed antibodies mediate hyperacute rejection. A positive lymphocyte cross-match means a high probability of hyperacute rejection, and the donor is rejected for that recipient.

## Virology

CMV +ve hearts are given preferentially to CMV +ve patients, and CMV –ve hearts are ideally given to CMV –ve patients.

## Size

Recipient and donor should be a similar size (see 🕮 p506).

# Donor preparation

## Management of the organ donor

In order to optimize donor organs, the donor should have invasive monitoring with a radial arterial line, central venous line, and urinary catheter, pulse oximetry and EKG monitoring. Hypotension, hypothermia, and diabetes insipidus frequently accompany brain death. In addition to continued mechanical ventilation, it is often necessary to carry out large-volume fluid resuscitation, active warming, and inotropic support.

## Organ procurement

Harvest of multiple organs is common: liver and heart teams both want to maximize the length of IVC they procure, the same if lungs are being harvested. Inotropic, volume, and respiratory support is continued until the thoracic viscera are excised.

***Prior to incision***

- A final check of donor identity, UNOS ID, organ donation consent, brain death certification, and serology is made.
- The donor chart is read for history and investigations. The CXR, EKG, cardiac cath, and echo are reviewed; donor hemodynamics are discussed with the anesthesiologist.
- The plan/timings discussed with the abdominal procuring teams, and confirmed with the recipient surgeon (to ensure arrival of donor heart coordinates with timing of recipient cardiectomy).

***Technique of donor cardiectomy***

- A midline incision from sternal notch to pubis is made.
- A median sternotomy is made with liberal use of bone wax and cautery to minimize bleeding, pericardium opened and hitched to the skin.
- The ***heart is carefully examined*** for evidence of trauma, regional wall motion abnormalities, including palpation for aortic or coronary calcification, and thrills.
- Findings are discussed with the recipient transplant surgeon and the heart is accepted, and timing of cross-clamping agreed.
- Once the abdominal teams have finished their preliminary dissection, the aorta is mobilized as far as the innominate artery and a cardioplegia purse-string placed, and the SVC is mobilized, avoiding the SA node.
- When all teams are ready ***30 000U of heparin*** iv is given centrally.
- The central line is withdrawn so that the ***SVC can be clamped*** as far distally as possible, the ***azygos ligated***, and the IVC is partially divided anteriorly: make sure you have adequate length as the liver team will try to take this up to the RA.
- The ***right superior pulmonary vein is opened*** to vent the left heart.
- The cardioplegia tubing is carefully de-aired, connected to the cannula, aorta is cross-clamped, and 1L ***cold crystalloid cardioplegia*** (Celsior low to moderate $K^+$, high sodium extracellular in adults, Del Nido in children taking care to avoid overdistension of the root) infused.
- The pericardial sling is filled with ice-cold saline: palpate the heart frequently to ensure root pressure adequate and heart decompressed.

- The *IVC incision is completed*, the *right and left pulmonary veins are divided* at the pericardial reflection (Fig. 11.6).
- The heart is pulled down, the *clamp removed and the aorta is divided* as high as possible (in paediatric transplants the aortic arch, innominate vein, and pericardium are taken), *pulmonary arteries are divided* at the pericardial reflections, and the *SVC divided* below the tie.
- Finally the connective tissue at the pericardial reflection behind the left atrium is divided, allowing the *heart to be removed* from the chest.
- The heart is immediately immersed in cold isotonic solution.
- It is *inspected carefully* for any injuries (e.g., laceration to coronary sinus) and congenital anomalies.
- The chambers are irrigated with cold isotonic solution and the heart placed in three serial sterile plastic bags containing cold saline and ice without air, and transferred to an ice chest.
- If *lungs harvested* vent heart through LA appendage instead of RSPV, divide PA at bifurcation, and divide LA leaving atrial cuff and pulmonary veins with lungs.

## Organ preservation

Preservation means ischemic times up to 6h may be tolerated, but the key to minimizing ischemic injury remains minimizing ischemic time. A variety of storage solutions are used at temperatures of 4–10°C. Two categories exist: intracellular solutions characterized by high $K^+$ and low $Na^+$, e.g. Bretschneider (HTK), or extracellular with low to moderate $K^+$ and high $Na^+$, e.g. St Thomas' cardioplegia solution. Intracellular solutions may offer better protection against cellular edema; extracellular solutions may reduce cellular damage associated with hyperkalemia.

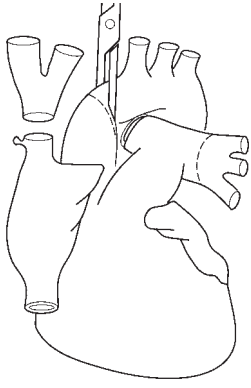


**Fig. 11.6** Donor heart harvest. Division of the great vessels: leave as much length on the heart as possible in all cases.

# Orthotopic cardiac transplant

## Key facts

There are several techniques of orthotopic heart transplantation:

- ***Original Shumway technique*** where donor LA is anastomosed to recipient LA, and RA to RA.
- ***Wythenshawe bicaval technique***: separate SVC to SVC and IVC to IVC anastomoses, instead of single right atrial anastomosis.
- ***Total technique***: bicaval end-to-end anastomoses, and bilateral pulmonary venous cuffs of recipient anastomosed to right and left half of donor LA.
- Anastomosing atria to atria is technically less demanding than bicaval anastomosis, or the total technique, but total technique reduces atrial redundancy and tricuspid regurgitation, improving atrial function, and reduces postoperative arrhythmias including need for PPM.

## Bicaval anastomosis technique

### *Recipient cardiectomy*

The recipient procedure is timed so that the recipient cardiectomy can be performed at the time of donor heart arrival: it is occasionally necessary to ask the procurement team to delay clamping the aorta if resternotomy and mediastinal dissection takes longer in complex reops.

- Midline sternotomy.
- Aortobicaval CPB with IVC and SVC cannulated directly and snared, and vacuum-assisted venous return.
- The *donor heart is inspected* before recipient cardiectomy proceeds.
- The *aorta is cross-clamped*.
- The *RA is incised* parallel to the AV groove, extending the incision towards the LA superiorly and medial to the IVC inferiorly.
- The heart is lifted up to expose the *LA* which *is incised* so that the incision extends towards the RA incision (Fig. 11.7).
- The *septum is divided* between the two incisions, and the LA incision continued in front of the left pulmonary veins and behind the LA appendage.
- The *aorta and pulmonary trunk are divided* just above the sinotubular junction and retracted inferiorly so that the superior aspect of the left atrium is exposed and the *atrial incisions continued* (Fig. 11.7).
- The *right atrium is divided* so that a generous cuff remains above the IVC cannula, and the SVC is divided far below the SVC cannula.

### *Donor heart implantation*

- *Left atrial anastomosis*: the donor and recipient left atria are anastomosed with continuous 3/0 Prolene, taking 1cm margins and everting so that intima is opposed to intima, starting at the LA appendage, finishing laterally over the pulmonary veins. This suture line must be hemostatic as it is relatively inaccessible once the transplant is completed.
- *Great vessels*: the donor and recipient *PA*s are then anastomosed end to end with 4/0 Prolene, followed by anastomosis of the *aorta*.
- To minimize ischemic time the aorta can be anastomosed immediately after the LA, and the clamp removed, allowing the heart to reperfuse while the right-sided anastomoses are completed.

- *Caval anastomoses*: the end-to-end anastomoses of the *IVC and SVC* are performed next with 4/0 Prolene: the IVC is first as this has the least possibility for adjustment, whereas both recipient and donor SVC are long enough to be tailored.
- *High-dose (1g) methylprednisone is given before reperfusion.*
- The heart is de-aired, the anastomoses secured, and after placing an aortic root vent, the cross-clamp is removed and the heart reperfused.

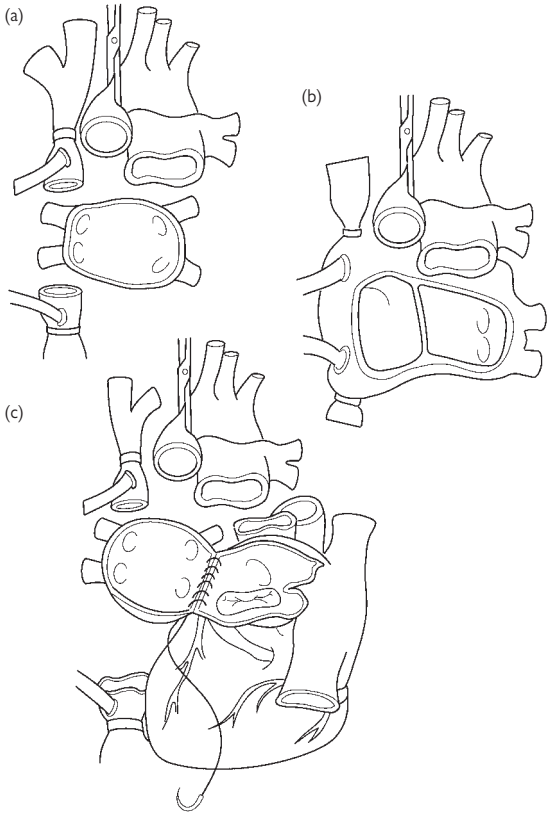



**Fig. 11.7** Orthotopic heart transplantation. (a) Recipient for bicaval anastomosis. (b) Recipient for classic anastomosis. (c) Left atrial anastomosis being performed during bicaval technique.

# Transplant care on ICU

## Key facts

- Transplant patients face the same problems and receive the same management as patients who have undergone cardiac surgery.
- ***Immunosuppression, rejection*** (📖 p514), and ***sepsis*** (including atypical organisms such as CMV and fungus) are specific to transplant patients.
- Complications more commonly seen in cardiac transplants than other cardiac surgery patients include ***sinus node dysfunction***, ***RV dysfunction*** which may be severe, and ***renal dysfunction***.

## Cardiac denervation

The donor heart is totally denervated. Resting sinus tachycardia (80–110bpm) is common because of loss of vagal parasympathetic input which normally has negative chronotropic effect. Epicardial pacing wires are routinely employed. Sinus node dysfunction can persist for several days: beyond that a permanent pacemaker is indicated.

## Right ventricular dysfunction

Myocardial dysfunction is associated with long periods of donor heart ischemia and reperfusion injury, which may be exacerbated by protamine administration. This results in RV dysfunction with tricuspid regurgitation, particularly in patients with preoperative pulmonary hypertension. It presents as a ***disproportionate rise in CVP*** compared to PA pressures, ***low cardiac output state*** (rising lactate, decreased urine output and $SVO_2$, hypotension, acidosis, and ventricular arrhythmias). Echocardiography is diagnostic. This requires aggressive management (including early recognition of cause of rising CVP): but it is best to avoid this by appropriate donor and recipient selection. Various agents dilate the pulmonary vasculature, reducing PVR: milrinone and inhaled NO have already been discussed (📖 p191). Prostaglandin E (30–50ng/kg/min) has been used in this context. RVAD support is required in refractory cases.

## Renal function

Renal failure is common after cardiac transplant: up to 15% of patients will require renal replacement therapy in the perioperative period. This is because firstly, there a number of common risk factors for ischemic cardiomyopathy and renal dysfunction (age, hypertension, diabetes); secondly because chronic hypoperfusion combines with the acute insult of surgery; and thirdly and importantly nephrotoxic drugs, particularly ciclosporin, are a mainstay of treatment of transplant patients. Renal management is the same as described for cardiac surgery. Additionally a rise in creatinine normally responds to lowering ciclosporin dosage, which can be changed for an alternative immunosuppressive regimen altogether.

## Immunosuppression

High-dose immunosuppression is given in the first few days postoperatively and reduced to maintenance doses. A standard regimen for the first 24h postoperatively includes ciclosporin 2mg/kg/day iv (or tacrolimus if

renal function is adequate), methylprednisolone 125mg iv three times daily, and azathioprine 2mg/kg iv 6-hourly, or mycophenolate mofetil 1.5gm iv 12-hourly. OKT3 may be used instead of ciclosporin.

## Antibiotic prophylaxis

As a result of early high-dose immunosuppression the patient is vulnerable to infection in the early postoperative period. Standard antibiotic prophylaxis lasts for 48h (flucloxacillin 1g 6-hourly iv). Nystatin or amphotericin are given orally for 4–6 weeks to prevent oral candidiasis. Trimethoprim 200mg 8-hourly po is given for prevention of *Pneumocystis jiroveci* (*carinii*) infection. CMV −ve blood products or leukocyte-depleted blood should be given to CMV −ve recipients. If the donor heart was CMV +ve then ganciclovir should be given for 2 weeks postoperatively.

### *Sepsis*

Common within the first 3 months, or after treatment of acute rejection. Commonest organisms are *E. coli*, *Pseudomonas* spp, *Staphylococcus* spp. Aggressive treatment with antimicrobials and reduction in immunosuppression may be required.

## Lung and heart–lung transplantation

In lung and heart–lung transplantation ventilatory management is the most significant challenge. This is primarily dependent on the quality of the donor organ, the duration of the ischemic time, and the quality of preservation. Respiratory dysfunction in the early postoperative period is the result of a number of processes:

- Ischemia/reperfusion injury as a result of long ischemic times and inadequate preservation leads to increased capillary permeability.
- Impairment of lymphatic drainage may exacerbate pulmonary edema.
- In bilateral sequential lung transplant the first implanted lung receives all the cardiac output while the second lung is being implanted: edema is normally more pronounced in this lung.
- Other causes of impaired respiratory function include subclinical disease in the implanted lung, shock, sepsis, multiple transfusion, and LV failure.

Positive pressure support is normally required for several days, but early extubation reduces the pressure on suture lines and negative effects on right ventricular function. Respiratory management aims to use the lowest $FiO_2$ to produce a $PaO_2$ >10kPa to minimize the risk of barotrauma. Peak inspiratory pressures are kept low by increasing respiratory rate rather than tidal volumes to achieve optimal minute volumes. Minute volumes are high to achieve relative hypocapnia which helps reduce pulmonary vascular resistance. PEEP is not used in patients with COPD in residual native lung as hyperinflation results. Great care must be taken to avoid volume overload and consequent pulmonary edema.

# Rejection

The goal of immunosuppression is to inhibit the immune response to alloantigen, while preserving the immune response to infection and malignancy. Immunosuppression consists of a short, intense induction phase followed by a permanent maintenance phase. A careful balance is maintained between therapeutic and toxic doses of immunosuppression: this is achieved by combining drugs with different mechanisms of action.

## Immunosuppressive agents

***Corticosteroids***

Corticosteroids inhibit the immune response at many levels. They decrease production of λ-interferon and interleukins which would normally cause upregulation of the lymphocyte response; and reduce macrophage function.

***Ciclosporin (CYA)***

Ciclosporin is a calcineurin inhibitor: it inhibits the production of IL-2 by T-helper cells, selectively reducing the cytotoxic T-cell response.

***Tacrolimus (FK506)***

Tacrolimus is also a calcineurin inhibitor with a similar profile.

***Mycophenolate mofetil (MMF)***

MMF inhibits purine synthesis in lymphocytes, reducing clonal expansion, and lymphocyte counts.

***Azathioprine (AZA)***

Azathioprine causes dose-related bone marrow suppression by suppression of purine synthesis.

***Sirolimus (rapamycin)***

Sirolimus stops IL-2 triggering clonal expansion of T lymphocytes.

***Daclizumub, basiliximab***

IL-2 receptor blockers which prevent clonal expansion of T cells.

***OKT3***

OKT3 is a monoclonal antibody produced in mice that binds the CD3 receptor site on cytotoxic T cells, preventing antigen recognition and clonal expansion.

***Polyclonal antibody (e.g., antithymocyte globulin)***

These are produced by animals after immunization with HLA: they attach to most circulating lymphocytes effecting a reduction in cell counts to less than 10% of normal.

***Simvastatin***

- Simvastatin has a moderate protective effect against chronic rejection: it suppresses T-cell function.

## Acute rejection

***Diagnosis***

Clinical features include low-grade fever, malaise, reduced exercise tolerance, pericardial rub, supraventricular arrhythmias, low cardiac output, and signs of congestive cardiac failure. Blood tests reveal a lymphocytosis. As all patients are on immunosuppression, symptoms may be minimal until rejection is quite advanced, so routine surveillance is undertaken. Hemodynamic measurements with a PA catheter may be helpful, but the gold standard diagnostic tool is RV endomyocardial biopsy: usually via a right internal jugular percutaneous approach. This is carried out every 7–10 days for a month, tapering to 2–6-month intervals. Lymphocyte infiltration and myocyte necrosis is used to grade the severity of cellular rejection. Biopsy-negative cardiac dysfunction raises the possibility of accelerated coronary artery disease or vascular rejection.

***Management***

- Asymptomatic mild rejection is monitored but not routinely treated.
- Myocyte necrosis is a definite indication for increased immunosuppression, but intermediate grades are treated according to clinical context.
- 3 days iv methylprednisolone 1000mg/day for acute rejection within 3 months of transplant, or for severe episodes.
- 100mg/day po methylprednisolone on a reducing dose over several weeks for all other episodes.
- Repeat endomyocardial biopsy after 7–10 days: repeat steroid course if no improvement, or give rescue therapy if hemodynamically unstable.
- Rescue protocols include methylprednisolone + OKT3, or methotrexate.
- Total lymphoid irradiation and photophoresis has been used in refractory rejection.
- Retransplantation is no longer performed in many centres for refractory rejection as results are extremely poor.

## Chronic rejection

Chronic rejection manifests as allograft CAD, detectable in over half of cardiac transplant patients within 5 years of transplantation, and within months in a few patients. Other processes that have been associated with allograft coronary disease include increased donor age, hyperlipidemia, and CMV infection. Ischemia is frequently silent as the heart is denervated. The clinical picture includes ventricular arrhythmias, congestive cardiac failure, and sudden death. Coronary angiography is the gold standard investigation but may underestimate the extent of the disease as the morphology is diffuse, smooth, intimal proliferation. There is no effective treatment apart from retransplantation. Prophylaxis centers on the same risk reduction strategies used in native ischemic heart disease.

# Heart transplantation results

## Infection

Non-specific suppression of lymphocyte activation, differentiation, proliferation, and function by immunosuppression means that transplant patients are particularly prone to infection which may be life threatening. The use of more selective immunosuppressive agents such as ciclosporin and tacrolimus, and a more conservative approach to treating low-grade, asymptomatic rejection, mean that the frequency and severity of infection post-transplant have fallen in the last few years (Table 11.1).

- CMV, *Toxoplasma gondii*, HBV, HCV, and HIV may be transmitted to the recipient from the graft.
- Infections in first month are most commonly nosocomial, months 1–6 opportunistic, after 6 months community acquired or opportunistic.

**Table 11.1** Infections post transplantation

| Infection/site | Organism | Treatment |
|---|---|---|
| **Within 30 days** | | |
| Pneumonia | Gram −ve bacilli | Imipenem, ceftazidime |
| Sternal wound | *Staphylococcus epidermidis*, *S. aureus*, Gram −ve bacilli | Flucloxacillin, vancomycin<br>Imipenem, ceftazidime |
| Line associated | *S. epidermidis*, *S. aureus*, Gram −ve bacilli | Flucloxacillin, vancomycin<br>Imipenem, ceftazidime |
| | *Candida albicans* | Fluconazole, |
| Urinary tract | Gram −ve bacilli, *Enterococcus* | Imipenem, ceftazidime |
| | *Candida albicans* | Fluconazole |
| Mucocutaneous | Herpes simplex virus, | Aciclovir |
| | *Candida sp.* | Fluconazole |
| **After 30 days** | | |
| Pneumonia | *Pneumocystis jiroveci (carinii)* | Trimethoprim |
| | Cytomegalovirus, herpes simplex virus | Ganciclovir, aciclovir |
| | *Haemophilus influenzae*, | Augmentin<br>Amphotericin |
| | *Cryptococcus, Aspergillus, Myocbacterium* spp. | Isoniazid, rifampicin |
| CNS | *Toxoplasma gondii, Aspergillus, Cryptococcus, Listeria* | Daraprim<br>Amphotericin |
| GI tract | *Candida albicans*, herpes simplex virus, *Aspergillus*, | Fluconazole<br>Aciclovir |
| Cutaneous | Herpes simplex virus, varicella-zoster, *Candida, Mycobacterium* | |

***Renal failure***

Cardiac and renal disease frequently coexist because they share common etiologies. Up to 70% of patients awaiting heart transplantation suffer from diabetes and atherosclerosis in addition to low cardiac output states, which predispose to renal impairment. Progression of renal failure is common in the later stages of cardiac disease: many patients develop end-stage renal failure while on the waiting list for heart transplantation. Native renal function may also be compromised after heart transplantation by factors including perioperative low cardiac output states, nephrotoxic antibiotics, and immunosuppressive regimens particularly calcineurin inhibitors which cause irreversible interstitial fibrosis. Although renal function can improve dramatically after restoration of blood flow, a minority of patients develop dialysis-dependent renal failure following transplantation. The survival of such patients is dramatically reduced: a creatinine clearance <40mL/min is therefore an accepted contraindication to heart transplantation. Frequent monitoring of ciclosporin levels with dose titration, and avoidance of hypovolemia are important adjuncts in the prevention of renal failure.

***Hypertension***

The majority of heart transplant recipients develop severe to moderate hypertension that is usually refractory to single therapy management. The etiology is poorly understood, but it is thought to include a combination of fluid retention as a result of ciclosporin-induced nephrotoxicity, and sympathetic-mediated vasoconstriction.

***Hyperlipidemia and glucose intolerance***

Hyperlipidemia and glucose intolerance develop in most patients as a result of immunosuppressive regimens. Both are treated aggressively to reduce the risk of early CAD. Simvastatin is used for both its lipid lowering effect and its ability to suppress T-cell function and the onset of chronic rejection.

***Malignancy***

The incidence of malignancy is 100× greater in heart transplant patients than in the general population: the majority of neoplasias are lymphoproliferative disorders and cutaneous carcinomas. The reason for this is this decrease in the T-lymphocyte response to EBV as a result of immunosuppression. In addition to conventional therapies for malignancy (chemotherapy, radiotherapy, and resection), immunosuppression may be reduced, and EBV treated with ganciclovir.

***Other complications***

Osteoporosis, avascular necrosis, biliary stasis and cholelithiasis, and obesity are common.

## Results of cardiac transplantation

- UK 30-day mortality is 4%.
- 1-year survival is 82%.
- 5-year survival is 75%.

# Heart–lung transplantation

## Key facts

- Heart–lung transplantation is not performed often: in 2008 only 73 such transplants were performed worldwide.
- This is because as results of single and double lung transplantation have improved, the indication for heart–lung transplant has reduced. At the same time pressure on donor organs means that organ blocs are usually divided to ensure maximum benefit to the greatest number of high-priority recipients.
- The ***domino procedure***, where a heart–lung transplant was performed for septic lung disease and the healthy explanted heart then transplanted into a second recipient, is now rarely performed.

## Indications

Patients are in end-stage cardiopulmonary disease with a life expectancy of 18–24 months. Congenital heart disease with secondary pulmonary hypertension or Eisenmenger syndrome (30%) is the commonest indication. Primary pulmonary hypertension with right-sided heart failure (cystic fibrosis, COPD, idiopathic pulmonary fibrosis) is the second commonest indication, but these cases are increasingly treated by lung transplant as native RV function often recovers.

## Contraindications

- Significant multisystem disease.
- Renal dysfunction.
- Malignancy.
- HIV, hepatitis B, hepatitis C with biopsy-proven liver disease.
- Infection with resistant respiratory flora.
- Mechanical ventilation.
- Relative contraindications include age >50, active extrapulmonary infection, peptic ulcer disease, continued smoking, high-dose steroids.

## Investigations and management of the recipient

The panel of investigations listed on 📖 p504 is carried out, additionally:

- Sputum for Gram stain, AFB and bacterial, mycobacterial, and fungal cultures.
- Thoracic measurements.
- Cystic fibrosis patients require specialist management including endoscopic maxillary antrostomies for sinus irrigation to decrease the bacterial load in the upper respiratory tract.

## Matching donor to recipient

Donor organs are matched to recipient on the basis of:

- Strict ABO compatibility.
- Donor and recipient height so that lung sizes are appropriate.
- HLA matching is not necessary but improved graft survival is associated with matching HLA-B, HLA-A, and HLA-DR loci.
- Lymphocyte toxicity screen <25% reactive.
- Negative donor–recipient cross-match if the screen is >25%.

## Technique of donor procurement

- The set-up is similar to that described for cardiectomy on 📖 p508.
- Median sternotomy.
- The inferior pulmonary ligaments are divided.
- The pericardium is opened and hitched to the skin, dissected off the ascending aorta, vena cavae, PA, and trachea, and excised to the hilum.
- Cardioplegia is infused into the aortic root, pulmonoplegia infused into the PA (📖 p508), and topical cold poured over the heart.
- The SVC is ligated as high up as possible, the IVC divided. And the left side of the heart vented through a cut in the LA appendage.
- The heart–lung bloc is dissected from the esophagus from diaphragm to carina, and the posterior hilar attachments divided.
- Lungs are inflated and the trachea stapled at least four rings above the carina, divided superiorly, and the heart–lung bloc is removed.

## Organ preservation

Ischemia and reperfusion injury is minimized by limiting ischemic time to a maximum of 6–8h, maintaining hypothermia at 0–10°C, intra- and extracellular cardioplegia and pulmonoplegia solutions, $PGE_1$ which reduces pulmonary vasoconstriction, and pretreatment with iv steroids.

## Heart–lung implantation

- Median sternotomy.
- Anterior pericardium is excised, the aorta and cavae are taped.
- The right phrenic nerve is separated from the hilum.
- CPB is instituted via high aortic cannula and direct caval cannulas.
- Left and right pneumonectomies are performed by dividing left and right inferior pulmonary ligaments, PA, and veins and main bronchi.
- The heart–lung bloc is placed in the chest, with the lungs passed under the phrenic nerve pedicles.
- The tracheal anastomosis is performed with continuous 3/0 Prolene, the caval and aortic anastomosis with continuous 4/0 Prolene.

## Postoperative management

Early postoperative management centres on maintaining a balance between adequate perfusion and gas exchange, while minimizing fluid load, cardiac work and barotraumas. Cardiovascular management and complications are very similar to those outlined on 📖 pp512–513.

## Results for heart–lung transplantation

- 1-year survival is 64%.
- 5-year survival is 49%.

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